L05 — study notes

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Lecture 5 — Repeated Games I: Cooperation · Map of the notes

Estimated unit. Built on 2026-10-16 from the study guide before the lecture. Reconcile with the slides when Aula/Lecture05.pdf arrives (/notes 05 –reconcile).

One claim. A repeated game can sustain behaviour that no one-shot equilibrium allows — but only if tomorrow can depend on today. With a known last period and a unique stage equilibrium it cannot: the end unravels the whole game. **Two stage equilibria** give a carrot and a stick, so the early rounds can cooperate. **With no last period** the repeated game manufactures its own carrot — cooperation forever supports itself — and grim trigger sustains it iff $\delta \geq (T-R)/(T-P)$, which is $1/4$ in Tadelis's Prisoner's Dilemma. **The folk theorem then says almost anything individually rational can be sustained by patient players** — which explains cooperation as a possibility, not as a prediction.

Three layers per topic:

Layer	What it means here
U — intuition	Gibbons §2.3 and Cabral §7.3: the gain-today-versus-loss-forever trade-off, δ as a continuation probability, renegotiation
C — course (examinable)	Tadelis Defs. 10.1–10.7, Props. 10.1–10.3, (10.3), (10.5), Theorem 10.1, and the worked examples on pp. 191, 197–200, 209–214
G — formal	MWG App. A to Ch. 12 (Nash reversion, monotonicity in δ, minimax folk theorem), Fudenberg–Maskin, Abreu's optimal punishments. **Named, not examinable** unless the lecture brings them

Reading order

Part I — finite horizons (§10.1)

1. L05-01-finite-repetition-unraveling — Def. 10.1, **Prop. 10.1** (p. 192) and its induction proof written out; the PD unravels after **one** fold or after 500; Gibbons's Figures 2.3.1–2.3.2. *Settles:* horizon length and δ are irrelevant when the stage NE is unique.
2. L05-02-two-stage-carrot-stick — Tadelis's two-stage example (p. 191): (M, m) then (R, r) is an SPE **iff** $\delta \geq 1/2$; the folded game's three equilibria; Gibbons's Figures 2.3.3–2.3.5 and **renegotiation**; and a T-round extension where **the last cooperative round binds and the floor is** $\delta > 1/3$.

Part II — infinite horizons (§10.2–10.3)

3. L05-03-discounting-histories-strategies — Defs. 10.2–10.4, δ **as a continuation probability** ($0.97^{500} = 2.4 \times 10^{-7}$), the derivation of $s(v, \delta, T)$, why the **average payoff** of a constant stream is the constant, histories and strategies, **Prop. 10.2**.
4. L05-04-grim-trigger-spe — Def. 10.5, grim trigger, **Prop. 10.3** (one-stage deviation), the two checks and **(10.3):** $\delta \geq 1/4$, Gibbons's $V = 4 + \delta V$, Cabral's $\alpha \geq 1/2$, and **the bootstrap**.
5. L05-05-critical-discount-factor — **(10.5)** derived, $\delta^* = (T - R)/(T - P) = g/(g + \ell)$, the **signed comparative statics** (the sucker payoff drops out), three readings (patience, interest rate up to 300%, expected duration), MWG's Nash reversion.

Part III — the folk theorem (§10.6)

6. L05-06-folk-theorem — convex combinations and hulls, **Theorem 10.1**, Gibbons's proof ($\delta \geq \max_i (d_i - x_i)/(d_i - e_i) < 1$), the **alternating path to** $(2, 2)$ with its three checks (**the bad period binds,** $\delta \geq 1/2$), a reading note on the Remark's arithmetic, Nash vs minimax thresholds, and **why "almost anything goes" is a selection problem**.

Support

7. L05-07-formulas — the specification mapped, every definition and number.
8. L05-08-pitfalls — 12 confusions with counterexamples, 22 original True/False items.
9. L05-09-python — every routine with `assert` checks against the books' numbers.

The one result to hold

$$\underbrace{R + \frac{\delta R}{1-\delta}}_{\text{cooperate forever}} \quad \geq \quad \underbrace{T + \frac{\delta P}{1-\delta}}_{\text{cheat once, punished forever}} \qquad \Longleftrightarrow \ \delta \ \geq \ \delta^* = \frac{T-R}{T-P}$$

A one-time gain $T - R$ against a per-period loss $R - P$ that lasts forever. In Tadelis's PD, $(5 - 4)/(5 - 1) = 1/4$. In general (10.5): $\delta \geq (v_i^d - v_i^c)/(v_i^d - v_i^*)$ for every player.

Notation. T is the PD temptation payoff in notes 04–06 and the companion, and the number of periods in notes 01–02 (Tadelis's $G(T, \delta)$). Player 1 chooses M/F (mum/fink), player 2 m/f, as in Tadelis and the Lecture 1 slides.

Numbers and cases to have memorized

Case	Number	Meaning
Finite PD, any T, any δ	fink every period	Prop. 10.1
Two-stage game, p. 191	$\delta \geq 1/2$	$+1$ today vs 2δ tomorrow
Grim trigger, Tadelis/Gibbons PD	$\delta^* = 1/4$	(10.3)
Grim trigger, Cabral	$\alpha \geq 1/2$	same gain, cheaper punishment
Alternating path to $(2, 2)$	$\delta \geq 1/2$; $(2.158,\ 1.842)$ at $\delta = 0.9$	the bad period binds
Folk region (Tadelis PD)	corners $(1, 1), (1, 4.6), (4, 4), (4.6, 1)$	feasible and $v_i > 1$

Figures

Built by `figures/make_figures.py` (run: `python Notas/L05/figures/make_figures.py`; ends with `assert` checks on the plotted numbers).

Figure	Shows	Lives in
L05-unraveling	the PD and the PD $+1$: same unique NE	L05-01-finite-repetition-unraveling §3
L05-two-stage-carrot-stick	the 3×3 game, its folded first stage at $\delta = 1/2$, gain vs loss	L05-02-two-stage-carrot-stick §2
L05-finite-coop-rounds	rounds of (M, m) in a 10-round SPE against δ	L05-02-two-stage-carrot-stick §4
L05-discounting	0.97^t on a log scale; $s(v, \delta, T) \to v/(1-\delta)$	L05-03-discounting-histories-strategies §2
L05-deviation-stream	cumulative PV of deviating minus cooperating at $\delta = 0.15, 0.25, 0.5$	L05-04-grim-trigger-spe §4.2
L05-grim-threshold	cooperate vs deviate average payoffs; $\delta^* = 1/4$ and $1/2$	L05-05-critical-discount-factor §2
L05-folk-theorem	feasible set, folk region, $(4.4, 2)$, alternating path	L05-06-folk-theorem §2

Interactive companions

- The Patience Threshold (`companions/companion-grim-trigger.html`) — PD payoffs T, R, P, S and δ; reads out $\delta^* = (T-R)/(T-P)$ (the answer), the present values of cooperating and deviating, the gain and the per-period loss, and the alternating path's averages and threshold; panels for the cooperate/deviate lines and for the feasible set with the folk-theorem region. Defaults are Tadelis p. 197; a Gibbons p. 90 preset. Linked from notes 04, 05, 06.
- The Endgame Reward (`companions/companion-finite-carrot-stick.html`) — δ, rounds T, temptation, carrot and stick payoffs of Tadelis's p. 191 game; reads out how many early rounds of (M, m) survive in an SPE, the reward-phase length, the one-round threshold $(1/2)$ and the floor $(1/3)$; panels for the round-by-round incentive check and rounds against δ. Linked from notes 01, 02.

Both are checked by `companions/check_companions.py` (Node runs each page's model; Python recomputes it).

Scope and ceiling

- **Examinable (C):** Tadelis §10.1–10.3 and §10.6 as listed; the one-stage deviation principle as Tadelis states it (Prop. 10.3); the threshold (10.5) and its PD form.
- **Named only (G):** MWG App. A to Ch. 12 (Nash reversion, Prop. 12.AA.2's monotonicity, the minimax folk theorem 12.AA.5); Fudenberg–Maskin; Abreu's optimal punishments; renegotiation-proofness.
- **Not here — Lecture 6:** tacit collusion (§10.4), reputation and the trust game (§10.5), Gibbons §2.3.C–D (Cournot collusion, efficiency wages), M&R Ch. 8 → Lecture 6.
- **Out of scope:** imperfect monitoring beyond the pointer to price wars, mechanism design, cheap talk, moral hazard (`CLAUDE.md` scope block).
- **The notes' own construction:** the T-round reward-phase extension of the two-stage game (L05-02-two-stage-carrot-stick §4, second companion) is not in the books; it is marked as such.

Course-wide position

Lecture	Relation to this one
L1 — dominance, IESDS	the PD's dominant strategy is why the one-shot game fails; the stage game is solved by dominance
L2 — Nash equilibrium	stage NE are the carrots, sticks and punishment payoffs v_i^*
L3 — backward induction, SPE	the unraveling is backward induction; SPE is the solution concept throughout
L4 — multistage games, bargaining	Props. 9.1–9.3, the Prisoner-Revenge Game, Theorem 9.1 (one-stage deviation); discounting
L5	**finite vs infinite repetition, grim trigger, (10.5), the folk theorem**
L6 — collusion, relational contracts	(10.5) applied to Cournot/Bertrand, price wars, reputation, efficiency wages
L7, L8 — incomplete information	reputation with private types is outside this unit

Rules pack: rules/game-theory (items 2–4, 14–15). Study guide: Map/study-guide (Lecture 5 block).

Bibliography actually used in these notes

Book	Where
Tadelis (2013) Ch. 10 §10.1–10.3, 10.6, pp. 190–214 (and §9.4–9.5, pp. 180–185)	the spine: every C-layer statement

Book	Where
Gibbons (1992) §2.3.A–B, pp. 82–102	U layer: two-stage games, renegotiation, Friedman's theorem and its proof, reservation payoffs
Cabral (2017) §7.3, pp. 172–174	the indefinitely repeated reading, $\alpha \geq 1/2$, relational contracts
MWG §12.D (p. 401) and **App. A to Ch. 12** (pp. 417–422)	G layer

Related project material. No list, quiz, solution or video exists for Lecture 5 yet. Practice by number and page (statements not reproduced): Tadelis Ch. 10 exercises 10.2 (p. 215), 10.3 (pp. 215–216), 10.5 (pp. 216–217), 10.6–10.7 (pp. 217–218), 10.9 (pp. 218–219), 10.12 (p. 219, after Lecture 6); Gibbons 2.10 (p. 134), 2.11–2.12 (p. 135); Muñoz-Garcia & Toro-Gonzalez Ch. 6 Ex. 1 (p. 274); MWG Exercise 12.AA.1 (p. 420).

10.1 · Finitely repeated games: a unique stage equilibrium unravels every period

Claim. Repeat a game with a **unique** Nash equilibrium a known, finite number of times and the future can discipline nothing: the last period must be the stage equilibrium whatever happened before, so the second-to-last period has no reward or punishment left to offer, and the argument walks back to period 1. **Five hundred rounds of the Prisoner's Dilemma have exactly one subgame-perfect equilibrium: fink, every period.**

Prev: L05-00-map · Next: L05-02-two-stage-carrot-stick · Map: L05-00-map

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1 · C — the object: a stage game played T times

Definition 10.1 (Tadelis, p. 190). *Given a stage-game G, $G(T, \delta)$ denotes the finitely repeated game in which the stage-game G is played T consecutive times, and δ is the common discount factor.*

A repeated game is **a multistage game in which every stage is the same game** (p. 190, the chapter opener). Everything Lecture 4 proved for multistage games therefore applies; the repetition only adds structure. Two ingredients are implicit and both matter:

Ingredient	Meaning	Where it bites
Perfect monitoring	the outcome of every stage is observed before the next begins (Gibbons's definition, p. 84)	strategies can condition on the history (note L05-03-discounting-histories-strategies §4)
A known last period T	common knowledge that stage T is the end	the unraveling below

Gibbons states the same object without discounting: $G(T)$ with payoffs **the sum of the T stage payoffs** (p. 84). With $\delta < 1$ or $\delta = 1$ the result below is identical — it never uses the size of δ.

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2 · C — Proposition 10.1

Proposition 10.1 (Tadelis, p. 192). *If the stage-game of a finitely repeated game has a unique Nash equilibrium, then the finitely repeated game has a unique subgame-perfect equilibrium.*

It is a special case of **Proposition 9.3** (p. 181): a finite multistage game whose stage-games each have a unique Nash equilibrium has a unique subgame-perfect equilibrium. **The unique SPE is the stage equilibrium played in every period, after every history.**

Gibbons's version (p. 84): *if the stage game G has a unique Nash equilibrium then, for any finite T, the repeated game $G(T)$ has a unique subgame-perfect outcome: the Nash equilibrium of G is played in every stage.*

2.1 · The proof, by induction from the end

Let a^* be the unique stage equilibrium with payoffs $u_i(a^*)$.

1. **Stage T.** Every subgame starting at T is a one-shot play of G: no future depends on the action. Each player best-responds to what the others do now, so play is a Nash equilibrium of G (Prop. 9.2, p. 181) — and there is only one, a^*. **This holds after every history.**
2. **Stage $T-1$, after any history.** Whatever is played now, stage T delivers a^*. Player i's payoff from action profile a at $T-1$ is therefore

$$u_i(a) + \delta\, u_i(a^*),$$

   and $\delta\, u_i(a^*)$ is **the same constant in every cell**. Adding a constant to every payoff does not change any best response (Tadelis fn. 3, p. 181), so the stage-$(T-1)$ game is strategically G itself: its unique equilibrium is a^*.
3. **Inductive step.** Suppose that from stage $t+1$ on, a^* is played after every history. Then at stage t the continuation adds $\sum_{s=t+1}^{T} \delta^{s-t} u_i(a^*)$, again the same in every cell; the stage-t game is G plus a constant, and its unique equilibrium is a^*.
4. By induction, a^* is played at every stage after every history. Uniqueness at each step gives uniqueness of the SPE. ■

   The engine is step 2: a continuation that does not depend on today's action is a constant, and a constant cannot reward or punish. Conditioning tomorrow on today needs **at least two different things tomorrow could be** — two continuation equilibria. Remove that and nothing is left.

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3 · Worked — the Prisoner's Dilemma unravels

Tadelis's stage game (p. 197; also the folk-theorem matrix, p. 210):

	m	f
M	4, 4	−1, 5
F	5, −1	1, 1

Stage T. F strictly dominates M for player 1: $5 > 4$ against m and $1 > -1$ against f; symmetrically f for player 2. The unique NE is (F, f) — "in fact it is the unique dominant strategy!" (p. 192).

Stage $T-1$. Stage T will be (F, f) whatever happens, worth $\delta \cdot 1$ to each player. The game at $T-1$ is the matrix above with δ added to every entry. With $\delta = 1$ (Gibbons's no-discounting case):

	m	f
M	5, 5	0, 6
F	6, 0	2, 2

F still dominates: $6 > 5$ and $2 > 0$. **Unique NE (F, f) again.** The same addition happens at $T-2$ (now $+\delta + \delta^2$), and so on to period 1.

Stage T: the stage game G

	m	f
M	4, 4	-1, 5
F	5, -1	**1, 1**

Stage $T-1$: G + continuation $(1,1)$

	m	f
M	5, 5	0, 6
F	6, 0	**2, 2**

Figure — folding back one stage adds the same continuation payoff to every cell, so the equilibrium cannot move. Source: Prop. 10.1, p. 192; matrix p. 197; the +1 is Gibbons's Figure 2.3.2 logic, p. 83.

The right table is the left table shifted by one unit everywhere. Every comparison a best response makes — 5 vs 4, 1 vs −1 — becomes 6 vs 5, 2 vs 0: same signs, same equilibrium.

3.1 · Gibbons's two-stage Prisoners' Dilemma (pp. 82–83)

Gibbons's Figure 2.3.1 uses L_i for "defect" and R_i for "cooperate":

	L_2	R_2
L_1	1, 1	5, 0
R_1	0, 5	4, 4

Played twice with summed payoffs, the second stage is (L_1, L_2) after every first-stage outcome, so the first stage is the game of **Figure 2.3.2**: add $(1,1)$ to every cell.

	L_2	R_2
L_1	2, 2	6, 1
R_1	1, 6	5, 5

Unique NE (L_1, L_2). **The unique subgame-perfect outcome is (L_1, L_2) in both stages; cooperation, (R_1, R_2), cannot be achieved in either stage** (p. 84).

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4 · U — why the result feels wrong, and why it is right

Tadelis (p. 192): two players facing 500 rounds "can use the future to discipline behavior and try to cooperate by playing mum in early periods." The intuition fails **because it is common knowledge that the game ends after 500 periods**: at $T = 500$ both must fink; at $T = 499$ no reward-and-punishment incentive is left; and so on.

What the intuition gets wrong is not "the future matters" — it is "the future can be made to depend on today." A threat to punish tomorrow is only credible if punishing is itself an equilibrium of tomorrow's game, and a promise to reward is only credible if rewarding is one. With a unique stage equilibrium there is exactly one thing tomorrow can be.

Feature	Matters for the result?
Length of the horizon ($T = 2$ or $T = 500$)	**No** — the induction does not care
Size of δ	**No** — δ only scales a constant
Uniqueness of the stage NE	**Yes** — see L05-02-two-stage-carrot-stick
A known last period	**Yes** — see L05-04-grim-trigger-spe

The same holds for the finitely repeated Cournot and Bertrand games of Tadelis §5.2: their stage equilibria are unique, so repetition changes nothing (p. 192).

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5 · G — the same argument in MWG

MWG §12.D (p. 401) runs the argument for the finitely repeated **Bertrand** duopoly: in period T prices equal cost whatever happened earlier, so period $T-1$ is "strategically speaking, at the last period", and backward induction delivers T repetitions of the static Bertrand outcome. Gibbons's footnote 13 (p. 84) extends the proposition to **dynamic** stage games: a unique backwards-induction (or subgame-perfect) outcome of G is played in every stage of $G(T)$. Not examinable beyond the statement.

Links

- Next: what changes with **two** stage equilibria — L05-02-two-stage-carrot-stick.
- What changes with **no last period** — L05-03-discounting-histories-strategies, L05-04-grim-trigger-spe.
- Backward induction and Prop. 9.3: Lecture 3, Lecture 4.
- Dominance in the stage game: Lecture 1.
- Rules: rules/game-theory item 4 ("Repeated games are not one-shot games").
- Interactive: The Endgame Reward (`companions/companion-finite-carrot-stick.html`) — set $r = p$ (the two stage equilibria pay the same) and every round collapses to stage play.
- Exercises: Tadelis 10.6(b) (p. 217) and 10.7(b) (pp. 217–218); Gibbons 2.12 (p. 135).

10.1 · Two stage equilibria: a carrot, a stick, and cooperation in the early rounds

Claim. When the stage game has **two Pareto-ranked Nash equilibria**, the last period can still reward or punish: play the good one if everyone behaved, the bad one if anyone cheated. That gap buys behaviour in earlier periods **that is not a stage equilibrium at all** — in Tadelis's game, (M, m) in period 1 is subgame-perfect **iff** $\delta \geq 1/2$. Over a longer horizon the early rounds are the easy ones, because they have the most future at stake.

Prev: L05-01-finite-repetition-unraveling · Next: L05-03-discounting-histories-strategies · Map: L05-00-map

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1 · C — Tadelis's two-stage example (p. 191)

The stage game, played twice with discount factor δ:

	m	f	r
M	4, 4	−1, 5	0, 0
F	5, −1	1, 1	0, 0
R	0, 0	0, 0	3, 3

Its pure Nash equilibria. Check each cell by best responses.

Cell	Player 1's alternatives	Player 2's alternatives	NE?
$(M, m) = (4, 4)$	F gives $5 > 4$	f gives $5 > 4$	**no**
$(F, f) = (1, 1)$	M gives -1, R gives 0	m gives -1, r gives 0	**yes**
$(R, r) = (3, 3)$	M, F give 0	m, f give 0	**yes**

Two equilibria, **Pareto ranked**: (R, r) beats (F, f) for both. (M, m) beats both but is not an equilibrium — the upper-left 2×2 block is the Prisoner's Dilemma of L05-01-finite-repetition-unraveling.

The strategies (p. 191):

- **Player 1:** play M in stage 1. In stage 2 play R if (M, m) was played in stage 1, and F otherwise.
- **Player 2:** play m in stage 1. In stage 2 play r if (M, m) was played in stage 1, and f otherwise.

Formally (fn. 1): $s_1 = (s_{11}, s_{12}(h_1))$ with $s_{11} = M$ and $s_{12}(h_1) = R$ if $h_1 = (M, m)$, F if $h_1 \neq (M, m)$. One first-stage action and **nine** contingent second-stage actions — a strategy names an action for every history, including those that never happen.

1.1 · Verifying subgame perfection, every step

Stage 2. After every first-stage outcome the prescribed pair, (R, r) or (F, f), is a stage Nash equilibrium. No one-stage deviation pays in any of the nine stage-2 subgames.

Stage 1. Fix player 2's strategy. Player 1's three options, each with its continuation:

Player 1 plays	Stage-1 payoff	Stage-2 outcome	Total v_1
M	4	(R,r): 3	$4+3\delta$
F	5	(F,f): 1	$5+\delta$
R	0	(F,f): 1	$0+\delta$

R is never better than M ($\delta < 4+3\delta$). M beats F iff

$$4+3\delta \;\geq\; 5+\delta \iff 2\delta \geq 1 \iff \boxed{\delta \geq \tfrac{1}{2}}$$

(fn. 2: "M is a best response if and only if $-1 \geq -2\delta$, or $\delta \geq \frac{1}{2}$"). Player 2 is symmetric.

> **Gain from deviating today:** $5-4=+1$**. Loss tomorrow: the switch from** (R,r) **to** (F,f) **costs** $3-1=2$**, worth** 2δ **today.** Cooperate iff $2\delta \geq 1$. Tadelis (p. 191): "both players will incur a loss of -2 in the second-period stage-game … The temptation to deviate … [is] the added payoff of $+1$."

---

2 · Worked — folding the second stage into the first

Replace stage 2 by its prescribed payoff: $+3\delta$ to both after (M,m), $+\delta$ to both after any other cell. At $\delta = 1/2$:

	m	f	r
M	5.5, 5.5	−0.5, 5.5	0.5, 0.5
F	5.5, −0.5	1.5, 1.5	0.5, 0.5
R	0.5, 0.5	0.5, 0.5	3.5, 3.5

Three pure equilibria of the folded game:

Cell	Check	Repeated-game outcome
(M,m)	$5.5 \geq 5.5$ (vs F), $5.5 \geq 0.5$ (vs R) — **a tie at exactly** $\delta = 1/2$	$((M,m),(R,r))$: **cooperation in stage 1**
(F,f)	$1.5 > -0.5$, $1.5 > 0.5$	$((F,f),(F,f))$
(R,r)	$3.5 > 0.5$	$((R,r),(F,f))$

Stage game: two NE, Pareto ranked

	m	f	r
M	4, 4	-1, 5	0, 0
F	5, -1	**1, 1**	0, 0
R	0, 0	0, 0	**3, 3**

Stage 1 folded back, $\delta = 1/2$

	m	f	r
M	**5.5, 5.5**	-0.5, 5.5	0.5, 0.5
F	5.5, -0.5	**1.5, 1.5**	0.5, 0.5
R	0.5, 0.5	0.5, 0.5	**3.5, 3.5**

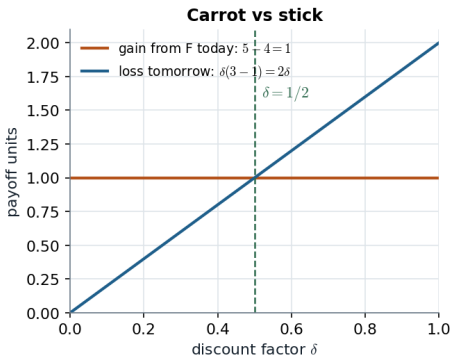


Figure — the stage game, its first stage after folding back at $\delta = 1/2$, and the gain-versus-loss comparison behind fn. 2. Source: Tadelis p. 191.

The middle table is where the work happens: the (M, m) cell gained $3\delta = 1.5$ while every other cell gained only $\delta = 0.5$, and that **extra 2δ is exactly enough** to offset the $+1$ from F. Below $\delta = 1/2$ the line on the right sits under the flat gain and (M, m) stops being an equilibrium of the folded game.

2.1 · Gibbons's version (Figures 2.3.3–2.3.4, pp. 84–86)

Gibbons's stage game (Figure 2.3.3) is the same idea with defect $= L$, cooperate $= M$, and the added equilibrium R: $(L, L) = (1, 1)$, $(M, M) = (4, 4)$, $(R, R) = (3, 3)$, $(L_1, M_2) = (5, 0)$. With summed payoffs ($\delta = 1$), adding $(3, 3)$ to the (M_1, M_2) cell and $(1, 1)$ to the other eight gives Figure 2.3.4, whose three pure equilibria correspond to the three subgame-perfect outcomes $((L_1, L_2), (L_1, L_2))$, $((R_1, R_2), (L_1, L_2))$ and $((M_1, M_2), (R_1, R_2))$ (p. 86). **The third is "qualitatively different": the first-stage outcome is not a Nash equilibrium of G.**

Gibbons's general point (p. 86): *if G has multiple Nash equilibria then there may be subgame-perfect outcomes of $G(T)$ in which, for any $t < T$, the outcome in stage t is not a Nash equilibrium of G.*

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3 · C — why this is Chapter 9 again

Tadelis (p. 191) points to the **Prisoner-Revenge Game** of §9.1 (p. 176): a reward-and-punishment second stage disciplines first-stage play. The difference is only that here the *same* game is repeated. The requirement, from Prop. 9.3 (p. 181) read in reverse:

> **Non-Nash behaviour early needs multiple continuation equilibria later** (p. 192). The good one is the **carrot**, the bad one the **stick**, and the players must be **patient enough** that the gap outweighs the one-period temptation.

---

4 · Extension — T rounds and a reward phase of length k

This section is the notes' construction on Tadelis's game, not the book's text. It is what the second companion computes.

Play (M, m) in rounds $1, \dots, n$ and (R, r) in rounds $n+1, \dots, T$ (the **reward phase**, of length $k = T - n$); any deviation sends play to (F, f) until the end. Write the payoffs generally: $a = 4$ on (M, m), $b = 5$ for F against m, $r = 3$ on (R, r), $p = 1$ on (F, f).

Rounds in the reward or punishment phase prescribe a stage equilibrium; no one-stage deviation pays there.

The last cooperative round, n. Deviating gains $b - a$ now; from round $n + 1$ the deviator gets p instead of r in each of the k remaining rounds:

$$\text{loss} = \sum_{j=1}^{k} \delta^j (r - p) = (r - p)\,\delta\,(1 + \delta + \cdots + \delta^{k-1}) = (r - p)\,\frac{\delta(1 - \delta^k)}{1 - \delta}.$$

So round n holds iff

$$\boxed{b - a \;\le\; (r-p)\,\frac{\delta(1-\delta^k)}{1-\delta}}$$

- $k = 1$: $b - a \le (r-p)\delta$, i.e. $\delta \ge \dfrac{b-a}{r-p} = \dfrac{1}{2}$ — Tadelis's condition.
- $k \to \infty$: the right side tends to $(r-p)\delta/(1-\delta)$, so **some** reward phase works iff

$$\delta > \frac{b-a}{(b-a)+(r-p)} = \frac{1}{1+2} = \frac{1}{3}.$$

At $\delta \le 1/3$ no finite game supports a single round of (M, m) with this profile.

Earlier cooperative rounds are slack (when $r \le a$). Let L_n be the loss at round n. The loss one round earlier is $L_{n-1} = \delta(a-p) + \delta L_n$, so

$$L_{n-1} - L_n = \delta(a-p) - (1-\delta)L_n = \delta(a-p) - (r-p)\delta(1-\delta^k) \;\ge\; \delta\big[(a-p) - (r-p)\big] \;\ge 0.$$

Every earlier round has more at stake, so **only the last cooperative round binds**, and the most cooperation is $n^* = T - k_{\min}$ with $k_{\min}$ the smallest k satisfying the box.

Worked, $T = 10$:

δ	$\dfrac{\delta(1-\delta^k)}{1-\delta}(r-p)$ for $k = 1, 2, 3, 4$	$k_{\min}$	rounds of (M, m)
0.30	0.60, 0.78, 0.83, 0.85 — never ≥ 1	—	**0**
0.34	0.68, 0.91, 0.99, 1.02	4	**6**
0.40	0.80, 1.12	2	**8**
0.50	1.00	1	**9**
0.90	1.80	1	**9**

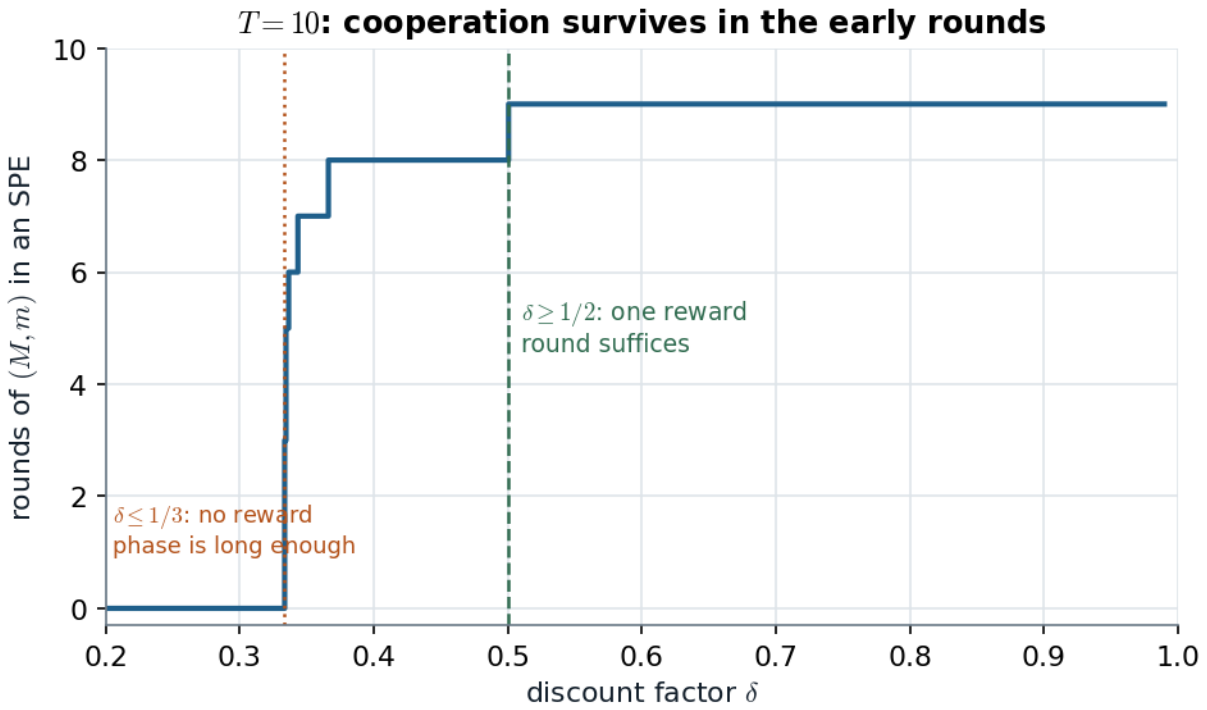


Figure — patience buys cooperative rounds, but never the last one. Source: the box above with Tadelis's payoffs, p. 191; computed by `coop_rounds` *in* `figures/make_figures.py`*.*

The curve never reaches 10: the last round must be a stage equilibrium (Prop. 9.2, p. 181). Between 1/3 and 1/2 the curve climbs in steps as one more reward round becomes necessary; from 1/2 on a single reward round is enough.

> **Interactive companion.** The Endgame Reward (`companions/companion-finite-carrot-stick.html`) — move δ, T and the carrot and stick payoffs; watch the bars (what a deviator loses) against the line (what he gains), and see how many early rounds of (M,m) survive.

---

5 · U/G — the renegotiation problem (Gibbons pp. 86–88)

The stick (F,f) hurts the punisher too. After a first-stage deviation both players would rather play (R,r), worth 3 each, than (F,f), worth 1: "bygones are bygones" and they might **renegotiate** (p. 87). If they would, the threat is empty, stage 2 is (R,r) after every history, and F is again a best response in stage 1.

Gibbons's fix (Figure 2.3.5, p. 87) adds two more equilibria, (P_1,P_2) with payoffs $(4,\frac{1}{2})$ and (Q_1,Q_2) with $(\frac{1}{2},4)$. Punish player 1's deviation with (Q_1,Q_2) and player 2's with (P_1,P_2); reward mutual cooperation with (R_1,R_2). Cooperating gives $4+3$, deviating at best $5+\frac{1}{2}$ (p. 88). Each punishment equilibrium is on the Pareto frontier of the stage equilibria, **so the punisher has nothing to renegotiate toward.** Named here as the **U/G** caveat; it is not part of Tadelis's treatment.

Links

- Previous: the unique-equilibrium case — L05-01-finite-repetition-unraveling.

- Next: infinite horizons manufacture the second equilibrium themselves — L05-03-discounting-histories-strategies, L05-04-grim-trigger-spe.
- Multistage games, the Prisoner-Revenge Game, Props. 9.1–9.3: Lecture 4.
- Rules: rules/game-theory items 3–4.
- Exercises: Gibbons 2.10 (p. 134), 2.11 (p. 135); Tadelis 9.1 (pp. 186–187) is the multistage warm-up (Lecture 4).

10.2 · Infinite horizons: discounting, average payoffs, histories and strategies

Claim. Remove the last period and the unraveling has nowhere to start. To compare infinite payoff streams we **discount** them, and δ can be read as **the probability that the relationship survives one more period** — so "infinitely repeated" means "no one knows which period is the last", not "lasts forever". The **average payoff** $(1-\delta)\times$ PV puts a stream on the same scale as a stage payoff, and a **strategy** is a rule mapping every past history to an action.

Prev: L05-02-two-stage-carrot-stick · Next: L05-04-grim-trigger-spe · Map: L05-00-map

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1 · C — present value (Definition 10.2)

Let $G(\delta)$ be the infinitely repeated game with discount factor δ (p. 193).

Definition 10.2 (p. 193). *Given the discount factor $0<\delta<1$, the present value of an infinite sequence of payoffs $\{v_i^t\}_{t=1}^{\infty}$ for player i is*

$$v_i = v_i^1 + \delta v_i^2 + \delta^2 v_i^3 + \cdots + \delta^{t-1} v_i^t + \cdots = \sum_{t=1}^{\infty} \delta^{t-1} v_i^t.$$

$\delta<1$ is **needed**: at $\delta=1$ a stream of 4s and a stream of 1s both sum to infinity and cannot be ranked (Gibbons, p. 89).

1.1 · Why the sum is finite (fn. 3, p. 193)

Suppose every stage payoff is below a bound $b>0$. Then $v_i < \sum_t \delta^{t-1} b \equiv B$. To see B is finite, multiply by $(1-\delta)$:

$$B(1-\delta) = \sum_{t=1}^{\infty} \delta^{t-1} b - \sum_{t=1}^{\infty} \delta^t b = b + \sum_{t=1}^{\infty} \delta^t b - \sum_{t=1}^{\infty} \delta^t b = b,$$

so $B = b/(1-\delta) < \infty$.

---

2 · U — δ as a continuation probability (pp. 193–194)

Suppose after each period the relationship continues with probability δ and ends with probability $1-\delta$. With no time discounting, the expected payoff is

$$Ev = v_i^1 + \delta v_i^2 + (1-\delta)\cdot 0 + \delta^2 v_i^3 + \delta(1-\delta)\cdot 0 + \cdots = \sum_{t=1}^{\infty} \delta^{t-1} v_i^t,$$

the same expression as Definition 10.2. The probability of playing infinitely many times is $\lim \delta^t = 0$: the game ends in finite time with probability 1, yet **at every period a future is still ahead** (p. 194).

$\delta = 0.97$	Value (p. 194)
Pr(at least 100 periods) $= 0.97^{100}$	0.0475

$\delta = 0.97$	Value (p. 194)
$\Pr(\text{at least 500 periods}) = 0.97^{500}$	2.4315×10^{-7} — "less than one in four million"

Both effects together (fn. 4, p. 194): with continuation probability p and time discount factor δ, the effective discount factor is the product $p\delta$. Gibbons writes the same thing as $\delta = (1-p)/(1+r)$, with p the per-stage end probability and r the interest rate (p. 90). Cabral calls this reading the **indefinitely repeated game** and notes that for risk-neutral players the two interpretations give the same computation (p. 174).

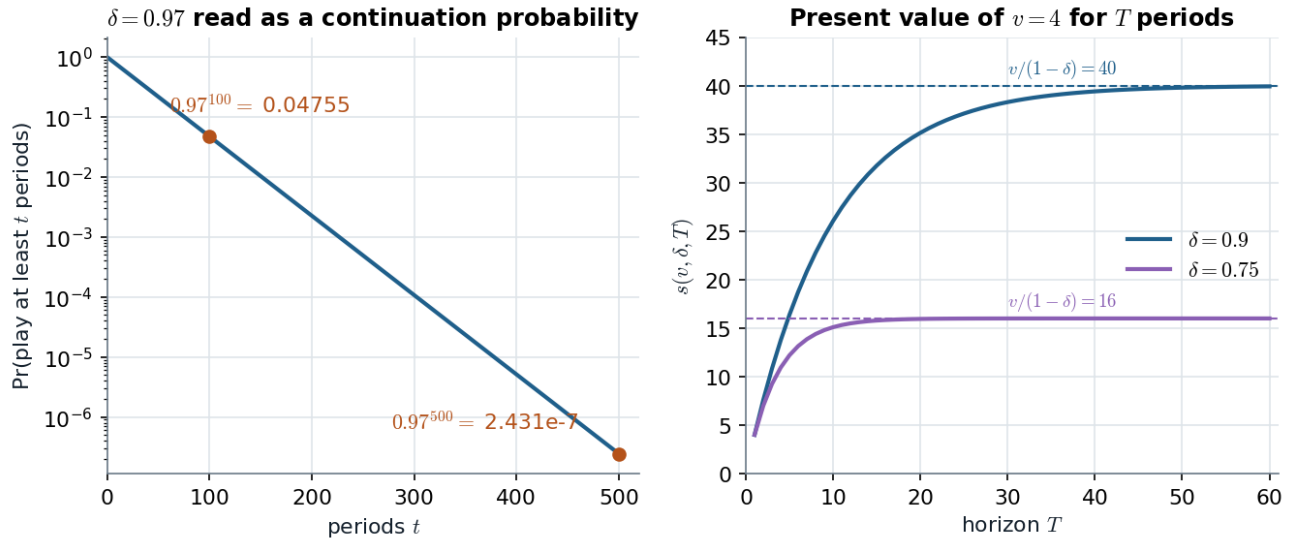


Figure — left, how unlikely a long game is at $\delta = 0.97$ (p. 194); right, $s(v, \delta, T)$ of (10.1) for $v = 4$ (the cooperative PD payoff, p. 197) converging to $v/(1-\delta)$.

The left panel is why "infinitely repeated" is not science fiction: even a 97% continuation probability makes 500 periods essentially impossible, yet the analysis below needs only that **some** future is always possible. The right panel shows that most of a discounted stream's value arrives early — at $\delta = 0.75$ the sum is within 1% of its limit by $T = 17$ ($0.75^{17} = 0.0075$).

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3 · C — the average payoff (Definition 10.3) and the sum $s(v, \delta, T)$

Definition 10.3 (p. 194). *Given $\delta < 1$, the average payoff of an infinite sequence of payoffs $\{v_i^t\}_{t=1}^{\infty}$ is*

$$\bar{v}_i = (1-\delta)\sum_{t=1}^{\infty} \delta^{t-1} v_i^t.$$

Why $(1-\delta)$. Take a constant payment v for T periods:

$$s(v, \delta, T) = v + \delta v + \delta^2 v + \cdots + \delta^{T-1} v. \tag{10.1}$$

Multiply by δ:

$$\delta s(v, \delta, T) = \delta v + \delta^2 v + \cdots + \delta^{T-1} v + \delta^T v. \tag{10.2}$$

Subtract (10.2) from (10.1): every term except the first of (10.1) and the last of (10.2) cancels,

$$s(v, \delta, T) - \delta s(v, \delta, T) = v - \delta^T v \quad \Longrightarrow \quad s(v, \delta, T) = \frac{v - \delta^T v}{1-\delta}.$$

Let $T \to \infty$; since $\delta^T \to 0$,

$$s(v,\delta) = \frac{v}{1-\delta}, \qquad \bar{v} = (1-\delta)\, s(v,\delta) = v.$$

> **The average payoff of a constant stream v is v itself.** That is the whole point of the normalization: average payoffs of repeated-game paths live on the **same axes as the stage game's payoffs**, which is what lets the folk theorem draw them in one picture (L05-06-folk-theorem).

Worked. (M, m) forever in Tadelis's PD: PV $= 4/(1-\delta)$ (at $\delta = 0.9$, 40); average $= (1-0.9)\cdot 40 = 4$. Maximizing the average or the PV is the same problem, since one is a positive multiple of the other (Gibbons, p. 97).

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4 · C — histories and strategies (Definition 10.4)

The game tree grows in length and in width: after each period, every action profile opens a new branch. In the repeated PD, **at stage 4 each player has 64 information sets** — one per possible 3-period history, $4^3 = 64$ (fn. 5, p. 195). In general, with $|A|$ action profiles in the stage game, there are $|A|^{t-1}$ histories before stage t.

Because **each information set is identified with a unique history**, a strategy can be written as a function of the history (p. 196).

Definition 10.4 (p. 196). *Let H_t denote the set of all possible histories of length t, $h_t \in H_t$, and let $H = \bigcup_{t=1}^{\infty} H_t$ be the set of all possible histories. A pure strategy for player i is a mapping $s_i : H \to S_i$ that maps histories into actions of the stage-game. Similarly a behavioral strategy of player i, $\sigma_i : H \to \Delta S_i$, maps histories into stochastic choices of actions in each stage.*

How big is a strategy? In the two-stage PD a strategy has **five** instructions — one for stage 1 and one for each of the four first-stage outcomes; in a two-stage game whose stage game is 3×3 (Gibbons's Figure 2.3.3, or Tadelis's p. 191 example) it has **ten** (Gibbons, p. 93). In $G(\delta)$ it is an infinite list. The *useful* strategies are short rules ("cooperate unless someone has ever defected") that compress the list.

Subgames. A subgame starts after any history and includes everything that follows. In $G(T)$ the subgame after stage t is $G(T-t)$; **in $G(\delta)$ every subgame is identical to the original game** (Gibbons, p. 94). This self-similarity is what makes the "bootstrap" argument of L05-04-grim-trigger-spe §5 consistent rather than circular.

> **The t-th stage on its own is not a subgame** (Gibbons, p. 95): a subgame includes all the moves that follow. Analysing stage t in isolation treats it as the last stage — exactly the mistake the repeated game exists to avoid.

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5 · C — the trivial equilibrium (Proposition 10.2)

Proposition 10.2 (p. 196). *Let $G(\delta)$ be an infinitely repeated game, and let $(\sigma_1^*, \ldots, \sigma_n^*)$ be a (static) Nash equilibrium strategy profile of the stage-game G. Define the repeated-game strategy for each player i to be the history-independent Nash strategy, $\sigma_i^*(h) = \sigma_i^*$ for all $h \in H$. Then $(\sigma_1^*(h), \ldots, \sigma_n^*(h))$ is a subgame-perfect equilibrium in the repeated game for any $\delta < 1$.*

Why. If the opponents' play ignores the history, today's action cannot change tomorrow; each period is a one-shot game in which σ_i^* is a best response to σ_{-i}^* (p. 196). So **fink forever is always an SPE of**

the repeated PD (p. 197) — the stick is available for free. The question of L05-04-grim-trigger-spe is whether anything *better* is.

Links

- Next: the grim-trigger equilibrium — L05-04-grim-trigger-spe.
- Discounting in multistage games and in Rubinstein bargaining: Lecture 4.
- Rules: rules/game-theory item 4.
- Exercises: Gibbons 2.12 (p. 135) asks for the definitions of a strategy and a subgame in a repeated game.

10.3 · Grim trigger: cooperation forever is subgame-perfect iff $\delta \geq 1/4$

Claim. In the infinitely repeated Prisoner's Dilemma, "cooperate until anyone defects, then defect forever" is a subgame-perfect equilibrium **iff** $\delta \geq 1/4$ with Tadelis's payoffs. The stick is the stage equilibrium repeated (always an SPE); the carrot is **the cooperative equilibrium itself** — a bootstrap, consistent because every subgame of $G(\delta)$ is the same game. Checking the infinitely many subgames reduces to **two** checks, one per state, thanks to the one-stage deviation principle.

Prev: L05-03-discounting-histories-strategies · Next: L05-05-critical-discount-factor · Map: L05-00-map

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1 · C — subgame-perfect equilibrium in a repeated game

Definition 10.5 (p. 196), in words: a profile $\sigma^* = (\sigma_1^*, \dots, \sigma_n^*)$ of history-contingent strategies is a **subgame-perfect equilibrium** of $G(\delta)$ if, **for every history** h, the strategies restricted to the subgame that starts after h form a Nash equilibrium of that subgame.

Gibbons (p. 95, after Selten 1965): *a Nash equilibrium is subgame-perfect if the players' strategies constitute a Nash equilibrium in every subgame.* Two consequences:

- **Off-path behaviour matters.** Histories that never occur in equilibrium (someone defected) are subgames too, and the threatened play there must be an equilibrium — a **credible** threat.
- **Infinitely many subgames** — one per history. Checking them one by one is impossible, which is why §3 matters.

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2 · C — the grim-trigger strategies (p. 198)

Tadelis's stage game (p. 197): $(M, m) = (4, 4)$, $(M, f) = (-1, 5)$, $(F, m) = (5, -1)$, $(F, f) = (1, 1)$.

- **Player 1:** in the first stage play $s_1^1 = M$. For any stage $t > 1$, play $s_1^t(h_{t-1}) = M$ **if and only if** the history h_{t-1} consists only of (M, m); otherwise play F.
- **Player 2:** the mirror strategy: m at $t = 1$, m iff $h_{t-1} = \{(M, m), \dots, (M, m)\}$, else f.

Read it as a two-state machine:

State	Entered when	Prescribed play	Leaves when
Cooperation	$t = 1$, or every past outcome was (M, m)	(M, m)	any outcome other than (M, m)
Punishment	some past outcome was not (M, m) — "be it once or more than once, be it by one player or both" (p. 198)	(F, f)	never

"Grim" because the punishment is **forever**; "trigger" because **any** deviation pulls it.

Gibbons's trigger strategy (p. 91) is the same with his labels: *Play R_i in the first stage. In the tth stage, if the outcome of all $t-1$ preceding stages has been (R_1, R_2) then play R_i; otherwise, play L_i.*

3 · C — the one-stage deviation principle (Proposition 10.3)

Proposition 10.3 (p. 198). *In an infinitely repeated game $G(\delta)$ a profile of strategies $\sigma^* = (\sigma_1^*, \dots, \sigma_n^*)$ is a subgame-perfect equilibrium if and only if there is no player i and no single history h_{t-1} for which player i would gain from deviating from $s_i(h_{t-1})$.*

It is the repeated-game form of **Theorem 9.1** (p. 185): *a one-stage unimprovable strategy must be optimal* (Lecture 4). A one-stage deviation changes the action at **one** history and follows the strategy everywhere else.

Why it survives an infinite horizon (fn. 6, p. 199). In a finite tree a profitable deviation can be broken into finitely many one-stage deviations, and induction finishes the proof. With an infinite horizon a profitable deviation may change infinitely many actions; but **discounting makes payoffs continuous at infinity**: changes after period N are worth at most a multiple of δ^N, so some deviation that differs from σ_i in only finitely many stages must also be profitable, and the finite argument applies (Fudenberg and Tirole, 1991, Thm 4.2, cited there).

> **The practical rule: list the states, and in each state check the single most tempting action today, followed by the equilibrium continuation from tomorrow.** For grim trigger that is two checks.

4 · Worked — the two checks

4.1 · Off the path: the punishment state

After any history with a deviation, the strategies prescribe (F, f) now and forever, whatever happens next. Facing f today and f forever after:

Player 1 plays today	Today	From tomorrow	Total
F (prescribed)	1	(F,f) forever: $\delta/(1-\delta)$	$1 + \dfrac{\delta}{1-\delta}$
M (deviation)	-1	still (F,f) forever: $\delta/(1-\delta)$	$-1 + \dfrac{\delta}{1-\delta}$

Deviating loses 2 today "with no gains in subsequent stages" (p. 199). **The punishment is credible for every** δ — it is Proposition 10.2's equilibrium.

4.2 · On the path: the cooperation state

After a history of only (M, m), with the opponent playing grim trigger:

Follow the strategy — 4 today, (M, m) forever after:

$$v_i = 4 + 4\delta + 4\delta^2 + \cdots = 4 + \frac{4\delta}{1-\delta}.$$

Deviate to F — 5 today, then the opponent plays f forever and F is the best reply:

$$v_i' = 5 + \delta + \delta^2 + \cdots = 5 + \frac{\delta}{1-\delta}.$$

No profitable deviation iff

$$4 + \frac{4\delta}{1-\delta} \;\geq\; 5 + \frac{\delta}{1-\delta}. \tag{10.3}$$

Solve, every step:

1. Multiply both sides by $1-\delta > 0$: $4(1-\delta) + 4\delta \geq 5(1-\delta) + \delta$.
2. Expand: $4 \geq 5 - 5\delta + \delta = 5 - 4\delta$.
3. Rearrange: $4\delta \geq 1$.
4. $$\boxed{\delta \geq \tfrac{1}{4}}$$

δ	v_i (cooperate)	v_i' (deviate)	Verdict
0.20	$4/0.8 = 5.000$	$5 + 0.2/0.8 = 5.250$	deviate: **not** an SPE
0.25	$4/0.75 = 5.333$	$5 + 0.25/0.75 = 5.333$	indifferent: SPE (weak)
0.50	8.000	6.000	cooperate: SPE

Two effects of today's action (p. 199): the **direct** effect on today's payoff (5 vs 4, +1) and the **indirect** effect on the continuation ($4\delta/(1-\delta)$ vs $\delta/(1-\delta)$, a loss of $3\delta/(1-\delta)$). Patience is what makes the indirect effect large.

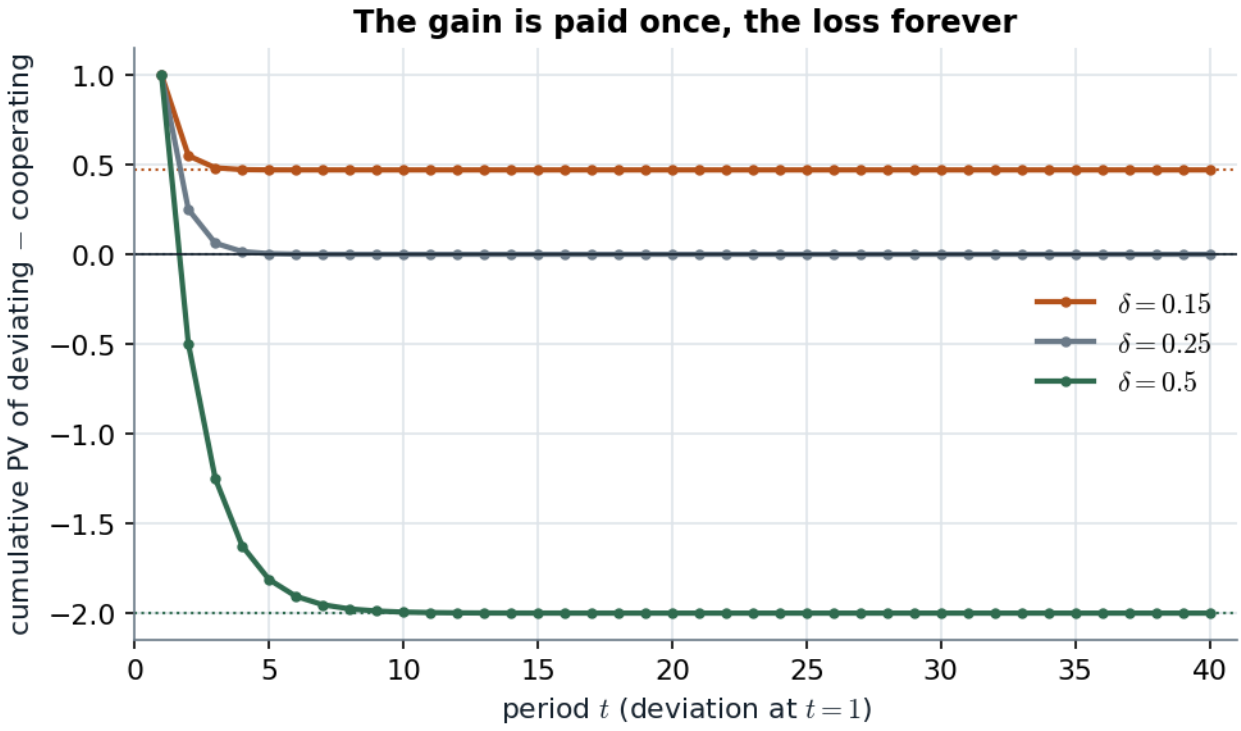


Figure — a deviation at $t=1$ pays +1 once and costs 3 in every later period; discounted, the running total converges to $1 - 3\delta/(1-\delta)$. Source: (10.3), pp. 199–200.

At $\delta = 0.25$ the running total converges exactly to zero — the threshold. Below it the one-period gain is never paid back; above it the losses overtake the gain within a few periods. The first loss already counts 3δ, so the verdict is mostly settled by the period right after the deviation.

Interactive companion. The Patience Threshold (`companions/companion-grim-trigger.html`) — move δ across δ^* and watch the verdict flip; move T, R, P and see which payoff moves the threshold and which (S) does not.

4.3 · Gibbons's recursive version (pp. 91–92)

Gibbons first shows grim trigger is a **Nash** equilibrium, with his PD $(L, L) = (1, 1)$, $(R, R) = (4, 4)$, (L_j vs R_i) $= 5$. Once someone defects, j's best reply is L_j forever. In a cooperative stage, let V be the value of making the L-or-R choice optimally now and whenever it recurs:

- if R_j is optimal, it yields 4 now and **the same choice next stage**: $V = 4 + \delta V$, so $V = 4/(1-\delta)$;
- if L_j is optimal: $V = 5 + \delta/(1-\delta)$.

R_j is optimal iff $\dfrac{4}{1-\delta} \geq 5 + \dfrac{\delta}{1-\delta}$ (2.3.1), i.e. $\delta \geq 1/4$. Then (p. 95) subgame perfection follows from two classes of subgames: after all-cooperation the continuation is again grim trigger, a Nash equilibrium of the whole game; after any defection it is (L_1, L_2) forever, also a Nash equilibrium. **Same threshold, with $S = 0$ instead of -1: the sucker payoff never enters** (it is not on either side of (10.3)).

4.4 · Cabral's indefinitely repeated version (pp. 172–173)

Cabral's game: (T, L) pays 5 each, the dominant-strategy equilibrium pays 4 each, and deviating from (T, L) pays 6; the game continues with probability α.

$$\frac{5}{1-\alpha} \;\geq\; 6 + \frac{4\alpha}{1-\alpha} \iff 5 \geq 6(1-\alpha) + 4\alpha \iff \alpha \geq \tfrac{1}{2}.$$

"A gain of 1 in the short run $(6-5)$ against a loss of 1 in every future period $(5-4)$" (p. 173).

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5 · U — the bootstrap

Tadelis (pp. 197–198): to support (M, m) forever we need a **stick** — (F, f) forever, an SPE by Prop. 10.2 — and a **carrot**. The carrot is the proposed equilibrium itself: if playing (M, m) forever is an SPE, then it is available as the reward continuation that supports itself.

> **"Indeed this is a circular argument, but it is not inconsistent"** (p. 197): a **bootstrap equilibrium**. It is consistent because every subgame of $G(\delta)$ is the same infinitely repeated game (L05-03-discounting-histories-strategies §4), so "cooperate forever from tomorrow" is exactly the object whose equilibrium status we are checking.

Back to Chapter 9 (p. 200). Think of "today" as stage 1 and "the infinite future" as stage 2. Non-Nash behaviour today needs (i) **multiple equilibria tomorrow** and (ii) **sufficient patience**. $\delta \geq 1/4$ is (ii). For (i), "from the unique equilibrium of the stage-game we get multiple equilibria of the repeated game" — the infinite horizon manufactures the carrot that the finite game of L05-01-finite-repetition-unraveling lacked.

The probability reading (p. 200): cooperation is credible "as long as the game is likely enough to be played again" — "what goes around comes around".

Links

- Previous: the objects used here — L05-03-discounting-histories-strategies.
- Next: the threshold for any payoffs, and what moves it — L05-05-critical-discount-factor.
- The folk theorem generalizes the construction — L05-06-folk-theorem.
- Theorem 9.1 and the Prisoner-Revenge Game: Lecture 4.
- The same construction for Cournot firms (§10.4) and for trust and reputation (§10.5): Lecture 6.
- Rules: rules/game-theory items 2 and 4.
- Exercises: Tadelis 10.2(a) (p. 215); Muñoz-Garcia & Toro-Gonzalez Ch. 6 Ex. 1 (p. 274); Gibbons 2.12 (p. 135).

10.3 · The critical discount factor: one formula, three readings

Claim. For any stage game, grim trigger supports a path paying v_i^c per period, against a best one-shot deviation v_i^d and a stage-Nash punishment v_i^*, **iff** $\delta \geq (v_i^d - v_i^c)/(v_i^d - v_i^*)$ **for every player**. In Prisoner's-Dilemma labels this is $\delta^* = (T-R)/(T-P)$: **the one-time gain over the one-time gain plus the per-period loss.** A bigger temptation or a softer punishment raises it; a better reward lowers it; the sucker payoff is irrelevant.

Prev: L05-04-grim-trigger-spe · Next: L05-06-folk-theorem · Map: L05-00-map

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1 · C — the general condition (10.5)

Tadelis's notation (p. 202): on the cooperative path player i earns v_i^c per period; the most profitable one-stage deviation earns v_i^d once; the stage Nash equilibrium used as the trigger punishment earns v_i^* per period. Player i does not deviate iff

$$\underbrace{v_i^c}_{\substack{\text{current payoff}\\ \text{from cooperate}}} + \underbrace{\frac{\delta v_i^c}{1-\delta}}_{\substack{\text{future payoff}\\ \text{if cooperate}}} \geq \underbrace{v_i^d}_{\substack{\text{current payoff}\\ \text{from best defect}}} + \underbrace{\frac{\delta v_i^*}{1-\delta}}_{\substack{\text{future payoff}\\ \text{if defect}}} .$$

"After some algebra" — here it is:

1. Multiply by $1-\delta > 0$: $v_i^c(1-\delta) + \delta v_i^c \geq v_i^d(1-\delta) + \delta v_i^*$.
2. Simplify the left side: $v_i^c \geq v_i^d - \delta v_i^d + \delta v_i^*$.
3. Collect δ: $\delta\,(v_i^d - v_i^*) \geq v_i^d - v_i^c$.
4. Divide by $v_i^d - v_i^* > 0$ (a deviation beats the punishment, since $v_i^d \geq v_i^c > v_i^*$):

$$\boxed{\delta \;\geq\; \frac{v_i^d - v_i^c}{v_i^d - v_i^*}} \tag{10.5}$$

Step 1 is also the **average-payoff** form: cooperate gives average v_i^c; deviate gives $(1-\delta)v_i^d + \delta v_i^*$, a weighted average of the deviation and the punishment.

With several players, each needs her own condition, so

$$\delta \geq \max_i \frac{v_i^d - v_i^c}{v_i^d - v_i^*} \quad \text{(Gibbons, Appendix 2.3.B, p. 101).}$$

Gibbons writes d_i, x_i, e_i for Tadelis's v_i^d, v_i^c, v_i^*. Since $d_i \geq x_i > e_i$, each fraction is **strictly below 1** (p. 102), so a patient enough $\delta < 1$ always exists — the seed of the folk theorem (L05-06-folk-theorem).

Lecture 6 applies (10.5) to asymmetric Cournot collusion, where the two firms need different thresholds (Tadelis §10.4, p. 202) → Lecture 6.

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2 · C — the Prisoner's Dilemma labels

Write the symmetric PD with the standard labels — **temptation T, reward R, punishment P, sucker** S, $T > R > P > S$ — so that $v^d = T$, $v^c = R$, $v^* = P$:

$$\delta^* = \frac{T-R}{T-P} = \frac{g}{g+\ell}, \qquad g = T-R \text{ (one-time gain)}, \quad \ell = R-P \text{ (per-period loss)}$$

Notation clash: T is also Tadelis's horizon in $G(T,\delta)$. In notes 04–06 and the companion T is the temptation payoff; in notes 01–02 it is the number of periods.

Source	(T,R,P,S)	g	ℓ	δ^*
Tadelis p. 197	$(5,4,1,-1)$	1	3	**1/4**
Gibbons p. 90	$(5,4,1,0)$	1	3	**1/4**
Cabral p. 173	$(6,5,4,\cdot)$	1	1	**1/2**

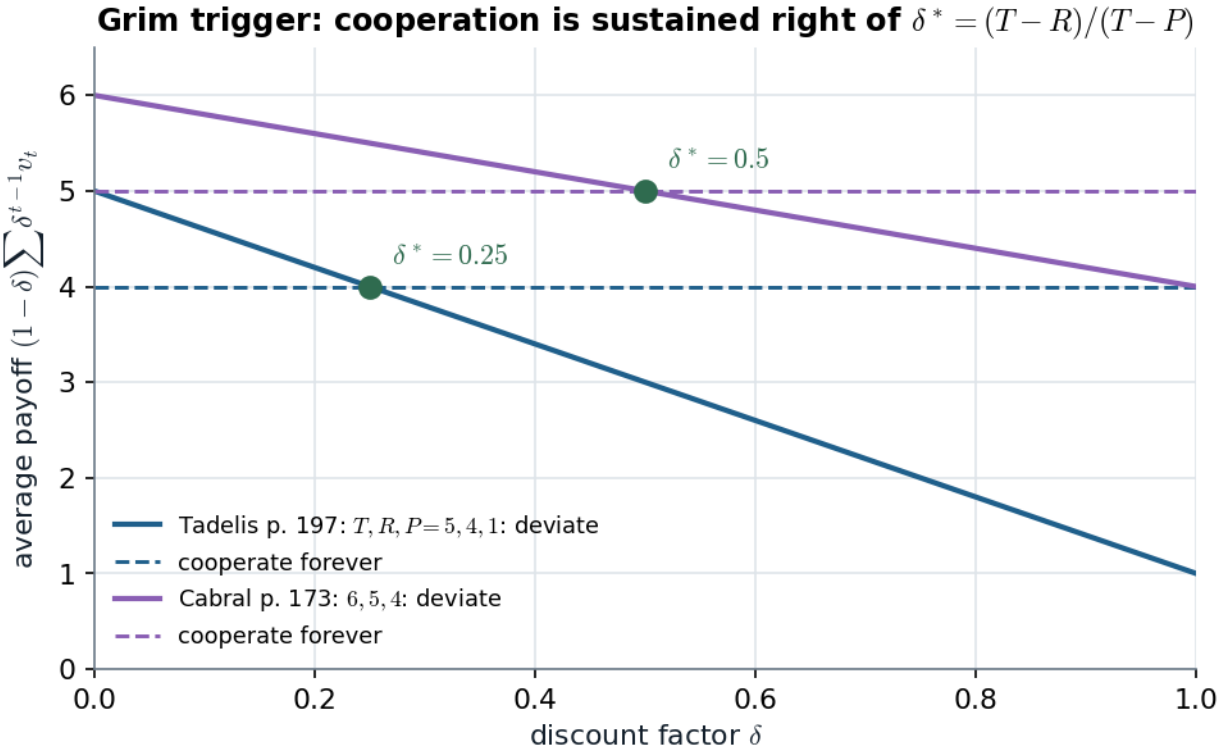


Figure — cooperate (dashed, flat at R) against deviate (solid, $(1-\delta)T + \delta P$); grim trigger is an SPE right of the crossing. Source: (10.3), p. 200; Cabral pp. 172–173.

The deviation line runs from T at $\delta = 0$ (only today counts) to P at $\delta \to 1$ (only the punishment counts). It crosses R where $(1-\delta)T + \delta P = R$, which is δ^*. Cabral's temptation is the same +1, but his punishment costs only 1 per period instead of 3, so his line is flatter and the crossing moves right, to 1/2.

> **Interactive companion.** The Patience Threshold (`companions/companion-grim-trigger.html`) — reproduce both lines (the "Gibbons p. 90" button gives the same 1/4 with $S = 0$), then move P toward R and watch δ^* approach 1.

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3 · C — comparative statics, with signs

From $\delta^* = (T-R)/(T-P)$:

$$\frac{\partial \delta^*}{\partial T} = \frac{(T-P)-(T-R)}{(T-P)^2} = \frac{R-P}{(T-P)^2} > 0, \qquad \frac{\partial \delta^*}{\partial R} = -\frac{1}{T-P} < 0,$$
$$\frac{\partial \delta^*}{\partial P} = \frac{T-R}{(T-P)^2} > 0, \qquad \frac{\partial \delta^*}{\partial S} = 0.$$

Payoff rises	δ^*	Why
Temptation T	**rises**	a bigger one-time gain needs more future to offset it
Reward R	**falls**	the gain shrinks and the loss grows
Punishment P	**rises**	a milder stick costs the deviator less per period
Sucker S	**unchanged**	on the grim-trigger path nobody is ever the sucker

The harshest credible punishment makes cooperation easiest. That is why grim trigger reverts to the *worst* stage equilibrium, and why Gibbons (p. 99) names Abreu's (1988) idea — punish with the worst subgame-perfect equilibrium, which can be harsher than stage-Nash reversion — as the route to cooperation when δ is not close to 1. In the PD the two coincide, because the stage equilibrium already gives each player her reservation payoff. **G**, named only.

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4 · U — three readings of the same number

(a) Patience. $\delta \geq \delta^*$ says the future weighs enough.

(b) Interest rate. With $\delta = 1/(1+r)$ (Gibbons, p. 89), $\delta \geq \delta^*$ iff

$$1 + r \leq \frac{1}{\delta^*} = \frac{g+\ell}{g} \iff r \leq \frac{\ell}{g}.$$

In Tadelis's PD $\ell/g = 3$: cooperation survives any per-period interest rate up to **300%**. The condition is generous because each period's loss is three times the one-time gain.

(c) Continuation probability. With δ the probability of another period (L05-03-discounting-histories-strategies §2), the expected number of periods is $1 + \delta + \delta^2 + \cdots = 1/(1-\delta)$. At $\delta^* = 1/4$ that is $4/3$: **an expected third of a period of future already disciplines a one-unit temptation** when each lost period costs 3.

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5 · G — MWG's version (Appendix A to Ch. 12)

MWG calls grim trigger **Nash reversion** (Definition 12.AA.1, p. 418): play a prescribed path until someone deviates, then the stage Nash equilibrium forever. Lemma 12.AA.1 (p. 418) is (10.5) for an arbitrary path, and the rewriting (12.AA.2) states it as "one-period gain $\leq$ discounted future loss". **Proposition 12.AA.2** (p. 420): *if a path can be sustained by Nash reversion at discount factor δ, it can be sustained for any $\delta' \geq \delta$* — the sustainable set only grows with patience, which is why δ^* is a threshold rather than a window. Not examinable unless the lecture brings it.

Links

- Previous: the derivation for Tadelis's numbers — L05-04-grim-trigger-spe.
- Next: what else patient players can support — L05-06-folk-theorem.
- Cournot and Bertrand collusion thresholds: Lecture 6.
- Rules: rules/game-theory item 4 ("the folk theorem needs sufficiently patient players and observable deviations").
- Exercises: Tadelis 10.2(a)–(b) (p. 215), 10.3 (pp. 215–216; milder "length-T" punishments), 10.5(c) (pp. 216–217), 10.9(b) (pp. 218–219); Muñoz-Garcia & Toro-Gonzalez Ch. 6 Ex. 1 (p. 274).

10.6 · The folk theorem: with enough patience, almost anything goes

Claim. Every payoff vector that is **feasible** (a weighted average of stage-game payoffs) and gives **each player more than a stage Nash equilibrium** can be approximated by a subgame-perfect equilibrium of $G(\delta)$ once δ is close enough to 1 (Theorem 10.1). Cooperation is one of a continuum of equilibria — so is perpetual defection, and so is a lopsided path that gives one player 4.6 and the other 1. **Repetition explains how cooperation *can* be sustained; it does not predict that it *will* be.**

Prev: L05-05-critical-discount-factor · Next: L05-07-formulas · Map: L05-00-map

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1 · C — feasible payoffs: convex combinations and the convex hull

Definition 10.6 (p. 209). *Consider two vectors $v = (v_1, \dots, v_n)$ and $v' = (v'_1, \dots, v'_n)$ in $\mathbb{R}^n$. We say that the vector $\hat{v} = (\hat{v}_1, \dots, \hat{v}_n)$ is a convex combination of v and v' if there exists some number $\alpha \in [0,1]$ such that $\hat{v} = \alpha v + (1-\alpha)v'$, or $\hat{v}_i = \alpha v_i + (1-\alpha)v'_i$ for all i.*

Worked (Figure 10.3, p. 210). $\alpha = 0.6$, $v = (4,4)$, $v' = (5,-1)$:

$$\hat{v} = 0.6(4,4) + 0.4(5,-1) = (2.4 + 2.0,\ 2.4 - 0.4) = (4.4,\ 2).$$

Definition 10.7 (p. 210). *Given a set of vectors $V = \{v^1, \dots, v^k\}$ in $\mathbb{R}^n$, the convex hull of V is*

$$\text{CoHull}(V) = \Big\{ v \in \mathbb{R}^n : \exists (\alpha_1, \dots, \alpha_k) \in \mathbb{R}^k_+,\ \sum_{j=1}^{k} \alpha_j = 1, \text{ such that } v = \sum_{j=1}^{k} \alpha_j v^j \Big\}.$$

Apply it to **average payoffs**. Repeating a pure profile s forever gives average payoff $u(s)$ (L05-03-discounting-histories-strategies §3), so the pure-profile payoff vectors are achievable averages, and their convex hull is the set of **feasible payoffs** (p. 211). For Tadelis's PD, $V = \{(4,4), (5,-1), (-1,5), (1,1)\}$ (Figure 10.4, p. 211).

Gibbons (p. 96) calls a payoff feasible if it is a convex combination of the pure-strategy payoffs, and notes how to *get* one exactly: a **public randomizing device** — flip a fair coin to choose between (L_1, R_2) and (R_1, L_2) and the expected payoff is $(2.5, 2.5)$. Without the device, Tadelis gets there **over time**, by alternating (§3).

---

2 · C — Theorem 10.1

Theorem 10.1 (The Folk Theorem) (p. 211). *Let G be a finite, simultaneous-move game of complete information, let $(v_1^*, \dots, v_n^*)$ denote the payoffs from a Nash equilibrium of G, and let $(v_1, \dots, v_n)$ be a feasible payoff of G. If $v_i > v_i^*$ for all $i \in N$, and if δ is sufficiently close to 1, then there exists a subgame-perfect equilibrium of the infinitely repeated game $G(\delta)$ that achieves an average payoff arbitrarily close to $(v_1, \dots, v_n)$.*

Gibbons's **Friedman (1971) Theorem** (p. 97) is the same statement with "achieves $(x_1, \dots, x_n)$ as the average payoff" (exact, with the randomizing device), e_i the Nash payoffs and x_i the target.

The intuition in four steps (pp. 211–212):

1. The target v is a weighted average of pure-profile payoffs (it is feasible).

2. With δ near 1, a deterministic **sequence** of pure profiles can mimic those weights, so its average payoff is (close to) v.
3. Because $v_i > v_i^*$ for everyone, reverting to the stage Nash equilibrium is a **punishment for every player** — the "wedge".
4. As $\delta \to 1$, even the largest one-period gain "is dwarfed by an infinite series of even small losses", so the wedge deters every deviation.

The proof for a stationary target (Gibbons, Appendix 2.3.B, pp. 100–102). Play the profile a_x that yields x; after any deviation, play the stage NE a_e forever. Let d_i be i's best one-shot deviation payoff against a_x, so $d_i \geq x_i > e_i$. By the computation of L05-05-critical-discount-factor §1, i conforms iff $\delta \geq (d_i - x_i)/(d_i - e_i)$; so the profile is a Nash equilibrium iff

$$\delta \;\geq\; \max_i \frac{d_i - x_i}{d_i - e_i} \;<\; 1.$$

Subgame perfection: after an all-a_x history the continuation is the same trigger profile (a Nash equilibrium of the game); after any other history it is a_e forever (also one). ■

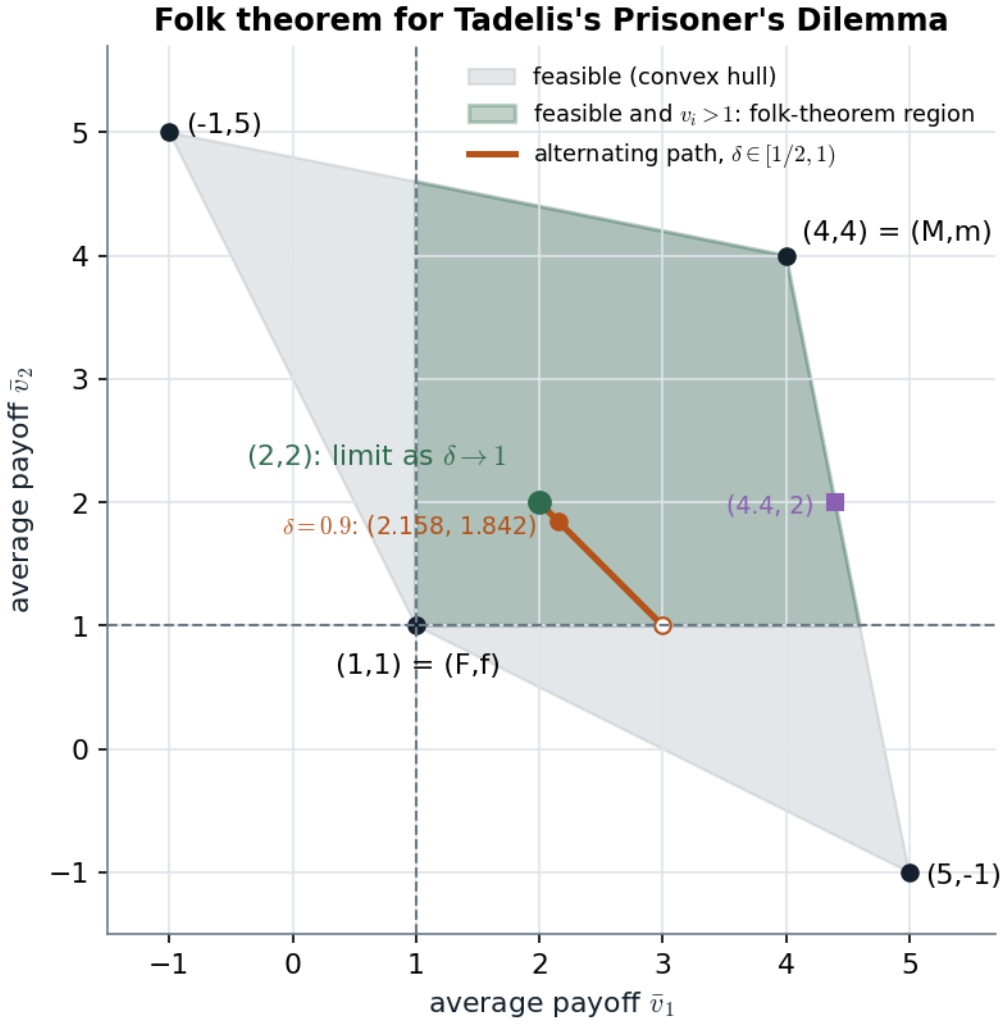


Figure — feasible set (grey), its strictly individually rational part (green), Definition 10.6's $(4.4, 2)$*, and the alternating path's averages for* $\delta \in 1/2, 1)$*. Source: Figures 10.3–10.5 and pp. 209–214.*

The green region's top-left corner is where the edge from $(-1, 5)$ to $(4, 4)$ — the line $v_2 = 5 - (v_1 + 1)/5$ — meets $v_1 = 1$: $v_2 = 5 - 2/5 = 4.6$. Every point in the green interior, including lopsided ones near

$(1, 4.6)$, is an SPE average for δ high enough. The orange segment starts at $(3, 1)$: at $\delta = 1/2$ player 2's average on the alternating path is exactly the punishment payoff 1, the edge of the region — which is why §3 finds $\delta \geq 1/2$.

> **Interactive companion.** [The Patience Threshold (`companions/companion-grim-trigger.html`) — the right panel redraws the feasible set and the folk-theorem region as T, R, P, S move, and plots the alternating path at the current δ.

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3 · Worked — reaching $(2, 2)$ by alternating (pp. 212–214)

$(2, 2) = \frac{1}{2}(5, -1) + \frac{1}{2}(-1, 5)$ — **not** a payoff of any pure profile, but feasible (Figure 10.5, p. 212).

Strategies (p. 212):

- **Player 1:** in period 1 play F. In even periods play M iff the pattern so far was $(F, m), (M, f), (F, m), ...$; otherwise F. In odd periods $t = 3, 5, ...$ play F.
- **Player 2:** in period 1 play m. In odd periods $t = 3, 5, ...$ play m iff the pattern was respected; otherwise f. In even periods play f.

On the path the outcomes alternate $(F, m), (M, f), (F, m), ...$, paying $(5, -1), (-1, 5), ...$

Average payoffs, every step. Split player 1's stream by parity:

$$\bar{v}_1 = (1-\delta)\big[5 + \delta(-1) + \delta^2(5) + \delta^3(-1) + ...\big] = (1-\delta)\big[5(1 + \delta^2 + \delta^4 + ...) + (-1)(\delta + \delta^3 + \delta^5 + ...)\big].$$

Both brackets are geometric series in δ^2: $1 + \delta^2 + \cdots = \dfrac{1}{1-\delta^2}$ and $\delta + \delta^3 + \cdots = \dfrac{\delta}{1-\delta^2}$. With $1 - \delta^2 = (1-\delta)(1+\delta)$,

$$\bar{v}_1 = (1-\delta)\,\frac{5-\delta}{(1-\delta)(1+\delta)} = \boxed{\frac{5-\delta}{1+\delta}}, \qquad \bar{v}_2 = \boxed{\frac{-1+5\delta}{1+\delta}} \text{ (same steps, roles swapped).}$$

As $\delta \to 1$: $\bar{v}_1 \to 4/2 = 2$ and $\bar{v}_2 \to 4/2 = 2$.

Subgame perfection: three kinds of history (one-stage deviation principle, Prop. 10.3):

State	Follow	Best one-stage deviation	Condition
Punishment ((F, f) forever)	1 per period	M or m: -1 today, punishment continues	always holds (loses 2)
Player 1's good period (he gets 5)	average $\dfrac{5-\delta}{1+\delta}$	M against m: 4 today, then 1 forever: $(1-\delta)4 + \delta$	holds for **all** $\delta \in (0, 1)$
Player 1's bad period (he gets -1)	average $\dfrac{-1+5\delta}{1+\delta}$	F against f: 1 today, then 1 forever: average 1	$\delta \geq \frac{1}{2}$

Good period, every step. Multiply $\frac{5-\delta}{1+\delta} \geq 4 - 3\delta$ by $1 + \delta$: $5 - \delta \geq (4 - 3\delta)(1 + \delta) = 4 + \delta - 3\delta^2$, i.e. $3\delta^2 - 2\delta + 1 \geq 0$. The discriminant is $4 - 12 < 0$, so the quadratic is always positive: **the condition always holds**.

Bad period, every step. $\frac{-1+5\delta}{1+\delta} \geq 1 \iff -1+5\delta \geq 1+\delta \iff 4\delta \geq 2 \iff \delta \geq \frac{1}{2}$.

Player 2's checks are the mirror images. **So the alternating path is an SPE iff $\delta \geq 1/2$, and its averages approach $(2,2)$ as $\delta \to 1$.**

> **The binding constraint is the bad period**: a player taking -1 today must be held there by the promise of 5 tomorrow. In general the bad period needs $\frac{S+\delta T}{1+\delta} \geq P$, i.e. $\delta \geq \frac{P-S}{T-P}$ — here the **sucker payoff matters**, unlike grim trigger on (R,R).

3.1 · The Remark: fine-tuning without $\delta \to 1$ (p. 214)

At $\delta = 0.9$ the alternating path gives $\bar{v}_1 = \frac{5-0.9}{1.9} = 2.158$ and $\bar{v}_2 = \frac{-1+4.5}{1.9} = 1.842$. Tadelis then alters single far-off periods to pull the averages toward $(2,2)$: a change at $t = 26$ moves them by 0.151 and 0.076 (to 2.007, 1.918), one at $t = 36$ by 0.013 and 0.066 (to 1.994, 1.984).

Reading note. Those four adjustments equal $\Delta \cdot \delta^{t-1}/(1+\delta)$ — e.g. $4 \times 0.9^{25}/1.9 = 0.151$ — which is the effect of changing the outcome **in period t and in every second period after it** ($(1-\delta)\Delta(\delta^{t-1} + \delta^{t+1} + ...)$). A one-off change in period 26 alone moves player 1's average by $(1-\delta)\delta^{25} \cdot 4 = 0.029$. The point of the Remark is unaffected: **far-off periods are a fine dial on the average**, so targets can be hit even at a fixed $\delta < 1$. (Verified in L05-09-python §6.)

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4 · U/G — what the threshold is: Nash payoffs or minimax payoffs

Tadelis's and Friedman's theorem punishes with a stage **Nash** equilibrium, so it covers payoffs above v_i^*. Gibbons (pp. 98–99) defines the **reservation payoff** r_i, the largest payoff i can guarantee whatever the others do; necessarily $r_i \leq e_i$. In the PD $r_i = e_i = 1$ (fink guarantees at least 1). In Cournot $r_i = 0 < e_i$ (only producing nothing guarantees anything).

- **Fudenberg and Maskin (1986):** for two players, r_i can replace e_i — any feasible payoff with $x_i > r_i$ is an SPE average for δ near 1 (Gibbons, p. 99).
- **MWG**: Prop. 12.AA.3 (p. 420) is the **Nash-reversion folk theorem**; the **minimax payoff** $\underline{v}_i$ (p. 421) is the lowest payoff the rivals can hold i to, payoffs above it are **individually rational**, and Prop. 12.AA.5 (p. 422), *the* Folk Theorem, supports every feasible individually rational payoff as $\delta \to 1$.

G, not examinable unless the lecture brings it. In Tadelis's PD the two versions draw the same green region.

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5 · U — what "almost anything goes" means for prediction

Tadelis's own word for the result is "rather disturbing" (p. 209), and the section title is "Almost Anything Goes".

What the folk theorem delivers	What it does not
a proof that cooperation **can** be self-enforcing among patient players	a prediction of **which** equilibrium is played
a whole region of SPE payoffs, efficient ones included	a reason to rule out $(1,1)$ — fink forever is still an SPE (Prop. 10.2)

What the folk theorem delivers	What it does not
conditions: patience and **observable** deviations	anything about settings where deviations cannot be seen (price wars, Tadelis p. 203 → Lecture 6)

"A prediction that does not select among equilibria is not a prediction" (rules/game-theory item 3). The folk theorem moves the question from *can they cooperate?* to *which of many equilibria will they coordinate on, and how?* — focal points, communication, institutions. Lecture 6's applications (tacit collusion, reputation, relational contracts) are cases where the context picks the equilibrium.

Links

- Previous: the threshold that the folk theorem drives to 1 — L05-05-critical-discount-factor.
- Support: L05-07-formulas, L05-08-pitfalls, L05-09-python.
- Applications of the same machinery: Lecture 6.
- Mixed strategies and Nash payoffs of the stage game: Lecture 2.
- Rules: rules/game-theory items 3, 4 and 14.
- Exercises: Tadelis 10.12 (p. 219; uses §10.5's trust game, so after Lecture 6), 10.9 (pp. 218–219); MWG Exercise 12.AA.1 (p. 420).

Lecture 5 · Formulas — every definition, every result, every number

Prev: L05-06-folk-theorem · Next: L05-08-pitfalls · Map: L05-00-map

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0 · The exam specification, mapped

The lecture has no slides yet; the specification is the study guide's Lecture 5 block and the chapter's own summary (§10.7, pp. 214–215). **Estimated** — reconcile when `Aula/Lecture05.pdf` arrives.

#	Item (study guide / Tadelis §10.7)	Note
1	Finitely repeated games; a unique stage NE gives a unique SPE "regardless of the discount factor's value" (Prop. 10.1)	L05-01-finite-repetition-unraveling
2	Multiple stage NE: carrot and stick support non-Nash play in early stages	L05-02-two-stage-carrot-stick
3	Infinite horizon: discounting, present value, average payoff, δ as continuation probability	L05-03-discounting-histories-strategies §1–3
4	Histories and strategies; Prop. 10.2	L05-03-discounting-histories-strategies §4–5
5	Grim trigger, the one-stage deviation principle (Prop. 10.3), (10.3)	L05-04-grim-trigger-spe
6	The threshold for any payoffs, (10.5), and what moves it	L05-05-critical-discount-factor
7	"The folk theorem teaches us that as the discount factor approaches 1 … almost anything can happen" (§10.7)	L05-06-folk-theorem
8	Application pointer: Cabral §7.3, relational contracts	L05-04-grim-trigger-spe §4.4; the rest in Lecture 6

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1 · Every definition and proposition

Result	Statement	Derived in
Def. 10.1 (p. 190)	$G(T,\delta)$: stage game G played T consecutive times, common discount factor δ	L05-01-finite-repetition-unraveling §1
Prop. 10.1 (p. 192)	unique stage NE $\Rightarrow$ unique SPE of $G(T,\delta)$: the stage NE every period	§2
Prop. 9.3 (p. 181)	the multistage version	§2

Result	Statement	Derived in
Two-stage example (p. 191, fn. 2)	(M,m) then (R,r), else (F,f): SPE iff $4+3\delta \geq 5+\delta \iff \delta \geq \frac{1}{2}$	L05-02-two-stage-carrot-stick §1
Notes' extension	T rounds, reward phase k: last cooperative round needs $b-a \leq (r-p)\dfrac{\delta(1-\delta^k)}{1-\delta}$; some k works iff $\delta > \dfrac{b-a}{b-a+r-p}$	§4
Def. 10.2 (p. 193)	$v_i = \sum_{t=1}^{\infty} \delta^{t-1} v_i^t$, $0<\delta<1$	L05-03-discounting-histories-strategies §1
fn. 3 (p. 193)	bounded payoffs: $\sum \delta^{t-1} b = b/(1-\delta)$	§1.1
fn. 4 (p. 194)	continuation probability p and discount δ: effective factor $p\delta$; Gibbons $\delta = (1-p)/(1+r)$ (p. 90)	§2
Def. 10.3 (p. 194)	$\bar{v}_i = (1-\delta)\sum_{t=1}^{\infty} \delta^{t-1} v_i^t$	§3
(10.1)–(10.2) (pp. 194–195)	$s(v,\delta,T) = \dfrac{v-\delta^T v}{1-\delta}$; $s(v,\delta) = \dfrac{v}{1-\delta}$; average of constant v is v	§3
Def. 10.4 (p. 196)	pure strategy $s_i : H \to S_i$; behavioural $\sigma_i : H \to \Delta S_i$	§4
Prop. 10.2 (p. 196)	history-independent stage NE is an SPE of $G(\delta)$ for any $\delta < 1$	§5
Def. 10.5 (p. 196)	SPE: Nash equilibrium after every history	L05-04-grim-trigger-spe §1
Grim trigger (p. 198)	M iff the history is all (M,m), else F forever	§2
Prop. 10.3 (p. 198)	SPE iff no player gains from a single deviation at a single history	§3
Thm 9.1 (p. 185)	a one-stage unimprovable strategy is optimal	§3
(10.3) (p. 200)	$4+\dfrac{4\delta}{1-\delta} \geq 5+\dfrac{\delta}{1-\delta} \iff \delta \geq \frac{1}{4}$	§4.2
Gibbons (2.3.1) (p. 92)	$\dfrac{4}{1-\delta} \geq 5+\dfrac{\delta}{1-\delta} \iff \delta \geq \frac{1}{4}$	§4.3
(10.5) (p. 202)	$\delta \geq \dfrac{v_i^d - v_i^c}{v_i^d - v_i^*}$	L05-05-critical-discount-factor §1
PD labels	$\delta^* = \dfrac{T-R}{T-P} = \dfrac{g}{g+\ell}$, $g = T-R$, $\ell = R-P$	§2
Signs	$\partial\delta^*/\partial T > 0$, $\partial\delta^*/\partial R < 0$, $\partial\delta^*/\partial P > 0$, $\partial\delta^*/\partial S = 0$	§3
Several players (Gibbons p. 101)	$\delta \geq \max_i \dfrac{d_i - x_i}{d_i - e_i} < 1$	§1

Result	Statement	Derived in
Interest-rate reading	$\delta = \dfrac{1}{1+r} \geq \delta^* \iff r \leq \ell/g$	§4
Def. 10.6 (p. 209)	convex combination $\hat{v} = \alpha v + (1-\alpha)v'$, $\alpha \in [0,1]$	L05-06-folk-theorem §1
Def. 10.7 (p. 210)	convex hull CoHull(V)	§1
Thm 10.1 (p. 211)	feasible v with $v_i > v_i^*$ $\forall i$, δ near 1 $\Rightarrow$ SPE with average arbitrarily close to v	§2
Alternating path (pp. 212–214)	$\bar{v}_1 = \dfrac{5-\delta}{1+\delta}$, $\bar{v}_2 = \dfrac{-1+5\delta}{1+\delta}$; SPE iff $\delta \geq \frac{1}{2}$	§3
General alternating bad period	$\dfrac{S+\delta T}{1+\delta} \geq P \iff \delta \geq \dfrac{P-S}{T-P}$	§3
G: MWG Def. 12.AA.1, Lemma 12.AA.1 (p. 418)	Nash reversion; its no-deviation test	L05-05-critical-discount-factor §5
G: MWG Prop. 12.AA.2 (p. 420)	sustainable at $\delta \Rightarrow$ sustainable at every $\delta' \geq \delta$	§5
G: MWG Props. 12.AA.3 (p. 420), 12.AA.5 (p. 422)	Nash-reversion folk theorem; minimax (individually rational) folk theorem	L05-06-folk-theorem §4
G: Fudenberg–Maskin (Gibbons p. 99)	two players: reservation payoffs r_i replace Nash payoffs e_i	§4

---

2 · Every number and worked case

2.1 · The games

Game	Payoffs	Source
Tadelis's PD	$(M,m) = (4,4)$, $(M,f) = (-1,5)$, $(F,m) = (5,-1)$, $(F,f) = (1,1)$; $T,R,P,S = 5,4,1,-1$	p. 197 (and p. 210)
Tadelis's two-stage game	PD block plus R/r: $(R,r) = (3,3)$, all other R/r cells $(0,0)$	p. 191
Gibbons's PD	$(L,L) = (1,1)$, $(L_1,R_2) = (5,0)$, $(R_1,L_2) = (0,5)$, $(R,R) = (4,4)$	pp. 82, 90
Gibbons's 3×3	adds $(R_1,R_2) = (3,3)$ to his PD (labels L,M,R)	p. 85
Cabral's PD	cooperate 5 each, deviate 6, Nash 4	pp. 172–173

2.2 · Thresholds

Case	Threshold
Two-stage, one reward round (p. 191)	$\delta \geq 1/2$
T-round extension, floor	$\delta > 1/3$; at $T = 10$: $\delta = 0.34 \to 6$ rounds, $0.4 \to 8$, $0.5 \to 9$
Grim trigger, Tadelis and Gibbons	$\delta^* = 1/4$
Grim trigger, Cabral	$\alpha \geq 1/2$
Alternating path, Tadelis	$\delta \geq 1/2$ (good period: every δ)

2.3 · Values

Quantity	Value	Source
0.97^{100}, 0.97^{500}	0.0475; 2.4315×10^{-7}	p. 194
Information sets at stage 4 of the repeated PD	$4^3 = 64$	fn. 5, p. 195
PV of 4 forever at $\delta = 0.9$	40; average 4	Def. 10.3
Grim trigger at $\delta = 0.2,\ 0.25,\ 0.5$	$(5,\ 5.25)$, $(5.333,\ 5.333)$, $(8,\ 6)$ (cooperate, deviate)	(10.3)
Convex combination	$0.6(4,4) + 0.4(5,-1) = (4.4, 2)$	p. 209
Folk region corners (Tadelis PD)	$(1,1), (1,4.6), (4,4), (4.6,1)$	Fig. 10.4
Alternating path at $\delta = 0.9$	$(2.158,\ 1.842)$	p. 214
Remark adjustments	$-0.151,\ +0.076$ $(t = 26)$; $-0.013,\ +0.066$ $(t = 36) \to$ $(1.994,\ 1.984)$	p. 214
Two-stage folded game at $\delta = 1/2$	$(M, m) = (5.5, 5.5)$, $(F, f) = (1.5, 1.5)$, $(R, r) = (3.5, 3.5)$	p. 191
Applications for Lecture 6	Cournot cutoffs 0.595 and 0.392 (§10.4, p. 202)	Lecture 6

Lecture 5 · Pitfalls and a True/False self-test

Prev: L05-07-formulas · Next: L05-09-python · Map: L05-00-map

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Part I — the confusions this material produces

1 · "A long finite game is almost an infinite one"

It is not, when the stage equilibrium is unique. **500 rounds and 2 rounds have the same unique SPE** (Prop. 10.1, p. 192): the induction never uses the length. What saves cooperation is the absence of a *known* last period (or a second stage equilibrium), not a large T. **Minimal counterexample:** the 500-round PD — fink in every period is the only SPE. → L05-01-finite-repetition-unraveling §4.

2 · "Finitely repeated games can never support non-Nash play"

Only with a **unique** stage NE. With two Pareto-ranked equilibria, (M, m) in stage 1 of Tadelis's two-stage game is subgame-perfect for $\delta \geq 1/2$ (p. 191). The last period, though, must always be a stage NE. → L05-02-two-stage-carrot-stick.

3 · Infinite horizon as "the game lasts forever"

δ can be a continuation probability; the game then ends in finite time with probability 1 (p. 194). What matters is that **at every period another period is possible**, so there is never a last period from which to unravel. → L05-03-discounting-histories-strategies §2.

4 · Nash vs subgame-perfect in the repeated game

A threat that is not an equilibrium of the subgame where it would be carried out is not credible. Grim trigger works because its punishment, (F, f) forever, *is* an SPE (Prop. 10.2). **Minimal counterexample:** a profile whose off-path behaviour is not a best response there can be a Nash equilibrium for high δ and an SPE for no δ — Tadelis Exercise 10.4 (p. 216) asks for exactly such a profile in the trust game of §10.5 (Lecture 6). The off-path plays are never reached, so Nash equilibrium does not test them; subgame perfection does.

5 · Checking only the equilibrium path

The one-stage deviation principle needs the check **in every state**, off the path included. For grim trigger the off-path check is trivial (-2 today, nothing gained), but in other profiles it binds: in the alternating path of §10.6 the binding constraint is the **bad period**, not the punishment. → L05-06-folk-theorem §3.

6 · Comparing today's gain with one period's loss

The loss from deviating is **every future period's** loss, discounted: $(R-P)\delta/(1-\delta)$, not $(R-P)\delta$. With Tadelis's numbers the first gives $\delta \geq 1/4$, the second would give $\delta \geq 1/3$. (The one-period version *is* right in the two-stage game, because there is only one future period.)

7 · Thinking the sucker payoff matters for grim trigger

On the grim-trigger path nobody is ever the sucker and the punishment is (F, f), so $\delta^* = (T-R)/(T-P)$ has no S in it: Tadelis ($S = -1$) and Gibbons ($S = 0$) both get $1/4$. S does matter for the **feasible set** and for the **alternating** path, whose bad period needs $\delta \geq (P-S)/(T-P)$.

8 · Reading the folk theorem as a prediction of cooperation

It says a whole region of payoffs is supportable — **including the non-cooperative** $(1, 1)$ and lopsided points like $(4.6, 1)$. It is an existence result about possibilities, and it raises the equilibrium-selection problem rather than solving it. → L05-06-folk-theorem §5.

9 · "Feasible" means "a payoff of some pure profile"

Feasible means a **convex combination** of those payoffs: $(2, 2)$ and $(4.4, 2)$ in the PD are feasible though no cell pays them. They are reached by alternating over time (Tadelis) or by a public randomizing device (Gibbons, p. 96).

10 · Mixing present values and average payoffs

Compare like with like: $4/(1-\delta)$ (PV) against $5 + \delta/(1-\delta)$ (PV), or 4 (average) against $(1-\delta)5 + \delta$ (average). Comparing 4 with $5 + \delta/(1-\delta)$ is the classic slip.

11 · The stage-t game is a subgame

It is not: a subgame includes all moves that follow (Gibbons, p. 95). Treating stage t as a one-shot game assumes it is the last one.

12 · Notation: T the horizon vs T the temptation

Tadelis writes $G(T, \delta)$ with T the number of periods; the PD label T is the temptation payoff. Context decides; these notes use T for the horizon in notes 01–02 and for the temptation in notes 04–06.

Part II — True/False self-test (original items)

1. In the Prisoner's Dilemma repeated 1,000 times with $\delta = 0.99$, there is a subgame-perfect equilibrium in which both players cooperate in the first period. **False.** The stage game has a unique NE, so by Prop. 10.1 the unique SPE plays (F, f) in every period; neither the horizon nor δ enters the argument.

2. In a finitely repeated game, behaviour in the last period of any subgame-perfect equilibrium must be a Nash equilibrium of the stage game. **True.** At T nothing in the future depends on today's action, so each player best-responds in the stage game (Prop. 9.2, p. 181) — whether the stage NE is unique or not.

3. In Tadelis's two-stage game (p. 191), the strategy profile that rewards (M, m) with (R, r) and punishes everything else with (F, f) is subgame-perfect for every $\delta \in (0, 1)$. **False.** Stage-2 play is a stage NE after every history, but stage 1 requires $4 + 3\delta \geq 5 + \delta$, i.e. $\delta \geq 1/2$.

4. If a stage game has two Nash equilibria that give every player the same payoff, repetition a finite number of times cannot support any non-Nash first-period behaviour through rewards and punishments. **True.** A carrot and a stick must differ in payoff; with $r = p$ the loss from deviating is zero and the one-period temptation is never offset.

5. Reading δ as the probability that the game continues, the infinitely repeated game ends in finite time with probability one. **True.** The probability of reaching period t is $\delta^{t-1} \to 0$ (p. 194); the analysis only needs a future to exist at each period.

6. The average payoff of the stream $(4, 4, 4, \dots)$ is $4/(1-\delta)$. **False.** That is its present value; the average payoff is $(1-\delta)\cdot 4/(1-\delta) = 4$ (Def. 10.3).

7. In the infinitely repeated PD, playing fink after every history is a subgame-perfect equilibrium only if δ is small. **False.** By Prop. 10.2 it is an SPE for every $\delta < 1$: history-independent play makes each period a one-shot game.

8. With Tadelis's payoffs, grim trigger is a subgame-perfect equilibrium of the infinitely repeated PD if and only if $\delta \geq 1/4$. **True.** Off the path (F, f) forever is an SPE; on the path $4+\frac{4\delta}{1-\delta} \geq 5+\frac{\delta}{1-\delta} \iff \delta \geq \frac{1}{4}$ (10.3).

9. To verify that grim trigger is subgame-perfect, it is enough to check that no player gains from a single deviation on the cooperative path. **False.** The one-stage deviation principle requires checking every history, including those after a deviation; grim trigger passes the off-path check, but it has to be made.

10. Raising the sucker payoff S of a Prisoner's Dilemma lowers the critical discount factor for grim trigger. **False.** $\delta^* = (T-R)/(T-P)$ does not contain S.

11. Making the stage Nash payoff P closer to the cooperative payoff R makes cooperation harder to sustain with grim trigger. **True.** $\partial\delta^*/\partial P = (T-R)/(T-P)^2 > 0$: a milder punishment needs more patience.

12. In Cabral's game, where cooperating pays 5, deviating 6 and the stage equilibrium 4, cooperation is sustainable by trigger strategies if the continuation probability is at least 1/2. **True.** $5/(1-\alpha) \geq 6 + 4\alpha/(1-\alpha) \iff \alpha \geq 1/2$ (pp. 172–173).

13. If two players have different critical discount factors under a grim-trigger profile, the profile is an SPE when the common δ exceeds the smaller one. **False.** Each player must be deterred: δ must exceed the **larger** threshold, $\max_i (d_i - x_i)/(d_i - e_i)$ (Gibbons, p. 101).

14. With $\delta = 1/(1+r)$, grim trigger in Tadelis's PD survives any per-period interest rate up to 300%. **True.** $\delta \geq 1/4 \iff 1 + r \leq 4 \iff r \leq 3$.

15. The payoff vector $(2, 2)$ is feasible in Tadelis's PD even though no pure profile yields it. **True.** $(2,2) = \frac{1}{2}(5,-1) + \frac{1}{2}(-1,5)$, a convex combination (Def. 10.7, Fig. 10.5).

16. By the folk theorem, the average payoff (0.5, 3) can be supported in Tadelis's repeated PD for δ close enough to 1. **False.** It is feasible, but player 1 gets less than the stage Nash payoff 1; Theorem 10.1 needs $v_i > v_i^*$ for every player.

17. In the alternating path $(F, m), (M, f), \dots$ of Tadelis's PD, the binding incentive constraint is a player's "bad" period. **True.** The good-period condition holds for all δ; the bad period needs $\frac{-1+5\delta}{1+\delta} \geq 1$, i.e. $\delta \geq 1/2$.

18. The folk theorem predicts that patient players in a repeated Prisoner's Dilemma will cooperate. **False.** It shows that cooperation is one of many SPE outcomes, alongside perpetual defection and lopsided paths; it does not select among them.

19. The one-stage deviation principle for infinitely repeated games relies on discounting. **True.** Discounting makes payoffs continuous at infinity, so a profitable infinite deviation implies a profitable finite one (fn. 6, p. 199).

20. In a two-stage game with stage-game equilibria worth $(3, 3)$ and $(1, 1)$, punishing with $(1, 1)$ is immune to renegotiation. **False.** After a deviation both players prefer $(3, 3)$; "bygones are bygones" undermines the threat (Gibbons, p. 87). Renegotiation-proof punishments need equilibria the punisher does not want to leave (Figure 2.3.5).

21. In the infinitely repeated game, every subgame is identical to the whole game. **True.** Gibbons (p. 94); it is what makes the bootstrap — the cooperative equilibrium serving as its own reward — consistent rather than circular.

22. Increasing the number of rounds T in Tadelis's three-action stage game can increase the number of rounds in which (M, m) is played in an SPE, but never to all T rounds. **True.** Longer games allow longer reward phases (useful when $1/3 < \delta < 1/2$), and the final round must be a stage NE, which (M, m) is not (L05-02-two-stage-carrot-stick §4).

Links

- Map: L05-00-map · Formulas: L05-07-formulas · Python: L05-09-python.
- Rules: rules/game-theory items 4, 14, 15; exam style rules/00_estilo_avaliacao.

Lecture 5 · Computation in Python

Every routine the lecture needs, each with `assert` checks against the books' own numbers. The blocks run top to bottom as one script (numpy and sympy only). **What was reused:** the project has dominance and IESDS code in `Listas/lista-lecture-01-check.py` but no Nash-equilibrium finder or repeated-game routine, so `pure_ne` and `coop_rounds` are written here and are the same functions the figures use (`figures/make_figures.py`).

Map: L05-00-map · Formulas: L05-07-formulas · Prev: L05-08-pitfalls

---

1 · Stage games and the unraveling → L05-01-finite-repetition-unraveling

```
from fractions import Fraction as Fr
import numpy as np
import sympy as sp

# Tadelis's Prisoner's Dilemma, p. 197: rows M, F (player 1); columns m, f (player
↪  2).
PD = {("M", "m"): (4, 4), ("M", "f"): (-1, 5), ("F", "m"): (5, -1), ("F", "f"): (1,
↪  1)}

def pure_ne(pay, tol=1e-12):
    """Pure Nash equilibria of a bimatrix {(row, col): (u1, u2)}; weak best responses
    ↪  count."""
    rows = sorted({r for r, _ in pay}); cols = sorted({c for _, c in pay})
    return sorted((r, c) for r in rows for c in cols
                  if all(pay[(r, c)][0] >= pay[(r2, c)][0] - tol for r2 in rows)
                  and all(pay[(r, c)][1] >= pay[(r, c2)][1] - tol for c2 in cols))

def fold(pay, cont):
    """Add a continuation payoff cont(cell) = (c1, c2) to every cell: one step of
    ↪  backward induction."""
    return {k: (v[0] + cont(k)[0], v[1] + cont(k)[1]) for k, v in pay.items()}

def unravel(pay, T, delta):
    """Backward induction in G(T, delta) when the stage NE is unique: the NE in every
    ↪  period.
    Returns the list of stage-NE sets, period T first."""
    out, carry = [], (0, 0)
    for t in range(T, 0, -1):
        g = fold(pay, lambda k: carry)            # continuation is the same in every
↪  cell
        ne = pure_ne(g)
        assert len(ne) == 1, "Prop. 10.1 needs a unique stage NE"
        out.append(ne)
        u = pay[ne[0]]
        carry = (delta * (u[0] + carry[0]), delta * (u[1] + carry[1]))
    return out
```

```
assert pure_ne(PD) == [("F", "f")]
assert all(ne == [("F", "f")] for ne in unravel(PD, 500, 0.97))    # 500 rounds, p.
 ↪  192
assert pure_ne(fold(PD, lambda k: (1, 1)))[0] == ("F", "f")        # Gibbons Fig.
 ↪  2.3.2 logic, p. 83
```

2 · Two stage equilibria: carrot and stick → L05-02-two-stage-carrot-stick

```
G3 = {("M", "m"): (4, 4), ("M", "f"): (-1, 5), ("M", "r"): (0, 0),
      ("F", "m"): (5, -1), ("F", "f"): (1, 1), ("F", "r"): (0, 0),
      ("R", "m"): (0, 0), ("R", "f"): (0, 0), ("R", "r"): (3, 3)}      # Tadelis p.
 ↪  191
assert pure_ne(G3) == [("F", "f"), ("R", "r")]

def stage1(delta):
    """First stage folded: (R,r) after (M,m), (F,f) otherwise (p. 191)."""
    return fold(G3, lambda k: (3 * delta, 3 * delta) if k == ("M", "m") else (delta,
     ↪  delta))

v_M = lambda d: 4 + 3 * d        # fn. 2, p. 191
v_F = lambda d: 5 + d
assert v_M(0.5) == v_F(0.5) and v_M(0.49) < v_F(0.49)
assert set(pure_ne(stage1(0.5))) == {("M", "m"), ("F", "f"), ("R", "r")}
assert ("M", "m") not in pure_ne(stage1(0.49))

def coop_rounds(d, T, b=5.0, a=4.0, r=3.0, p=1.0):
    """Most rounds of (M,m) in T repetitions: (M,m) x n, then (R,r), deviation ->
     ↪  (F,f).
    Checks every cooperative round (one-stage deviation principle). Same code as the
 ↪  figures."""
    best = 0
    for n in range(T):
        if all(b - a <= sum(d ** (s - t) * ((a if s <= n else r) - p) for s in
         ↪  range(t + 1, T + 1)) + 1e-12
               for t in range(1, n + 1)):
            best = n
    return best

def k_min(d, b=5.0, a=4.0, r=3.0, p=1.0, kmax=200):
    """Shortest reward phase that holds the last cooperative round: closed form of
     ↪  the box in L05-02 §4."""
    for k in range(1, kmax):
        if b - a <= (r - p) * d * (1 - d ** k) / (1 - d) + 1e-12:
            return k
    return None

for d, n in [(0.3, 0), (0.34, 6), (0.4, 8), (0.5, 9), (0.9, 9)]:
    assert coop_rounds(d, 10) == n
    assert n == 0 or n == 10 - k_min(d)                    # only the last round
     ↪  binds
```

```
assert k_min(1 / 3 - 1e-9) is None and k_min(0.3334) is not None   # floor delta >
 ↪ 1/3
```

3 · Discounting and average payoffs → L05-03-discounting-histories-strategies

```
def pv(stream, delta):
    """Definition 10.2 truncated to the given stream."""
    return sum(delta ** t * v for t, v in enumerate(stream))

def average(stream, delta):
    """Definition 10.3: (1 - delta) x PV."""
    return (1 - delta) * pv(stream, delta)

def s_finite(v, delta, T):
    """(10.1)-(10.2): v + delta v + ... + delta^(T-1) v = (v - delta^T v)/(1 -
     ↪ delta)."""
    return (v - delta ** T * v) / (1 - delta)

assert abs(s_finite(4, 0.9, 30) - pv([4] * 30, 0.9)) < 1e-12
assert abs(s_finite(4, 0.9, 10_000) - 40) < 1e-9 and abs(average([4] * 10_000, 0.9) -
 ↪ 4) < 1e-9
assert abs(0.97 ** 100 - 0.0475) < 1e-4 and abs(0.97 ** 500 - 2.4315e-7) < 1e-10   #
 ↪ p. 194
assert 4 ** 3 == 64                                  # information sets at stage 4,
 ↪ fn. 5, p. 195
```

4 · Grim trigger → L05-04-grim-trigger-spe

```
def grim_threshold(T, R, P):
    """(10.3)/(10.5) in PD labels: delta* = (T - R)/(T - P), exact if given
     ↪ ints/Fractions."""
    return Fr(T - R, T - P) if all(isinstance(x, int) for x in (T, R, P)) else (T -
     ↪ R) / (T - P)

def on_path_check(T, R, P, delta, horizon=4000):
    """One-stage deviation on the cooperative path, computed from explicit
     ↪ streams."""
    follow = pv([R] * horizon, delta)
    deviate = pv([T] + [P] * (horizon - 1), delta)
    return follow >= deviate - 1e-9

def off_path_check(P, S, delta, horizon=4000):
    """In the punishment state: F against f (P forever) vs M today (S) then P
     ↪ forever."""
    return pv([P] * horizon, delta) >= pv([S] + [P] * (horizon - 1), delta)

assert grim_threshold(5, 4, 1) == Fr(1, 4)                       # Tadelis p. 200;
 ↪ Gibbons p. 92
assert grim_threshold(6, 5, 4) == Fr(1, 2)                       # Cabral p. 173
assert on_path_check(5, 4, 1, 0.25) and not on_path_check(5, 4, 1, 0.24)
```

```
assert all(off_path_check(1, -1, d) for d in (0.01, 0.5, 0.99))  # punishment
 ↪  credible for every delta
# the table in L05-04 §4.2
for d, (vc, vd) in {0.2: (5.0, 5.25), 0.25: (16 / 3, 16 / 3), 0.5: (8.0,
 ↪  6.0)}.items():
    assert abs(4 / (1 - d) - vc) < 1e-12 and abs(5 + d / (1 - d) - vd) < 1e-12
# the deviation stream converges to 1 - 3 delta/(1 - delta)  (figure
 ↪  L05-deviation-stream)
assert abs(pv([1] + [-3] * 4000, 0.5) - (1 - 3 * 0.5 / 0.5)) < 1e-9
```

5 · The general threshold and its comparative statics → L05-05-critical-discount-factor

```
T_, R_, P_, S_, d_ = sp.symbols("T R P S delta", real=True)
vd, vc, vs = sp.symbols("v_d v_c v_star", positive=True)
# (10.5): solve v_c/(1-d) >= v_d + d v*/(1-d) for the boundary delta
boundary = sp.solve(sp.Eq(vc / (1 - d_), vd + d_ * vs / (1 - d_)), d_)[0]
assert sp.simplify(boundary - (vd - vc) / (vd - vs)) == 0

ds = (T_ - R_) / (T_ - P_)
signs = {x: sp.simplify(sp.diff(ds, x)) for x in (T_, R_, P_, S_)}
assert sp.simplify(signs[T_] - (R_ - P_) / (T_ - P_) ** 2) == 0     # > 0 when R > P
assert sp.simplify(signs[R_] + 1 / (T_ - P_)) == 0                  # < 0
assert sp.simplify(signs[P_] - (T_ - R_) / (T_ - P_) ** 2) == 0     # > 0 when T > R
assert signs[S_] == 0                                               # the sucker
 ↪  payoff is irrelevant

def threshold_many(players):
    """Gibbons App. 2.3.B: delta >= max_i (d_i - x_i)/(d_i - e_i), each < 1 when d_i
     ↪  >= x_i > e_i."""
    vals = [(d - x) / (d - e) for d, x, e in players]
    assert all(v < 1 for v in vals)
    return max(vals)

assert threshold_many([(5, 4, 1), (6, 5, 4)]) == 0.5
g, l = 5 - 4, 4 - 1
assert abs(1 / (1 + l / g) - 0.25) < 1e-12        # r <= l/g = 3 is the interest-rate
 ↪  reading
```

6 · The folk theorem → L05-06-folk-theorem

```
def alternating(T, R, P, S, delta):
    """(F,m),(M,f),... : averages (T + dS)/(1+d), (S + dT)/(1+d) and the three checks
     ↪  (pp. 212-214)."""
    v1, v2 = (T + delta * S) / (1 + delta), (S + delta * T) / (1 + delta)
    good = v1 >= (1 - delta) * R + delta * P - 1e-12      # deviate to M in one's
 ↪  good period
    bad = v2 >= P - 1e-12                                 # deviate to F in one's bad
 ↪  period
    return v1, v2, good and bad
```

```
v1, v2, ok = alternating(5, 4, 1, -1, 0.9)
assert ok and abs(v1 - 2.158) < 5e-4 and abs(v2 - 1.842) < 5e-4           # Remark,
 ↪ p. 214
assert alternating(5, 4, 1, -1, 0.5)[2] and not alternating(5, 4, 1, -1, 0.49)[2]
assert all(alternating(5, 4, 1, -1, d)[0] >= (1 - d) * 4 + d for d in
 ↪ np.linspace(0.01, 0.99, 99))
assert abs(alternating(5, 4, 1, -1, 1 - 1e-9)[0] - 2) < 1e-8               # -> (2,
 ↪ 2)
# series check of the closed form
assert abs(average([5, -1] * 3000, 0.9) - (5 - 0.9) / 1.9) < 1e-9

# Remark's adjustments: book = Delta * d^(t-1)/(1+d); a one-off change = (1-d) *
 ↪ d^(t-1) * Delta
book = lambda D, t, d=0.9: D * d ** (t - 1) / (1 + d)
assert [round(book(4, 26), 3), round(book(2, 26), 3), round(book(1, 36), 3),
 ↪ round(book(5, 36), 3)] == \
       [0.151, 0.076, 0.013, 0.066]
every_second = lambda D, t, d=0.9: (1 - d) * D * sum(d ** (t - 1 + 2 * j) for j in
 ↪ range(5000))
assert abs(book(4, 26) - every_second(4, 26)) < 1e-12
assert round((1 - 0.9) * 0.9 ** 25 * 4, 3) == 0.029

def in_hull(pt, V):
    """Point in the convex hull of a convex polygon V listed counter-clockwise (cross
     ↪ products)."""
    return all((b[0] - a[0]) * (pt[1] - a[1]) - (b[1] - a[1]) * (pt[0] - a[0]) >=
     ↪ -1e-12
               for a, b in zip(V, V[1:] + V[:1]))

HULL = [(1, 1), (5, -1), (4, 4), (-1, 5)]            # counter-clockwise, Figure 10.4
assert in_hull((2, 2), HULL) and in_hull((4.4, 2), HULL) and not in_hull((4.5, 4.5),
 ↪ HULL)
assert np.allclose(0.6 * np.array([4, 4]) + 0.4 * np.array([5, -1]), [4.4, 2])   #
 ↪ Def. 10.6
assert abs(5 - (1 + 1) / 5 - 4.6) < 1e-12             # IR corner (1, 4.6)
print("L05 python checks pass")
```

7 · The companions' own check

`companions/check_companions.py` runs each page's model under Node at the default, low-end, high-end and preset vectors, compares every read-out with an independent Python implementation, sweeps every control, and syntax-checks the page. Run: `python Notas/L05/companions/check_companions.py`.

Links

- Figures script (same `pure_ne`, `coop_rounds`): `figures/make_figures.py`.
- Notes served: L05-01-finite-repetition-unraveling · L05-02-two-stage-carrot-stick · L05-03-discounting-histories-strategies · L05-04-grim-trigger-spe · L05-05-critical-discount-factor ·

L05-06-folk-theorem.

- Earlier-lecture code: `Listas/lista-lecture-01-check.py` (dominance, IESDS).